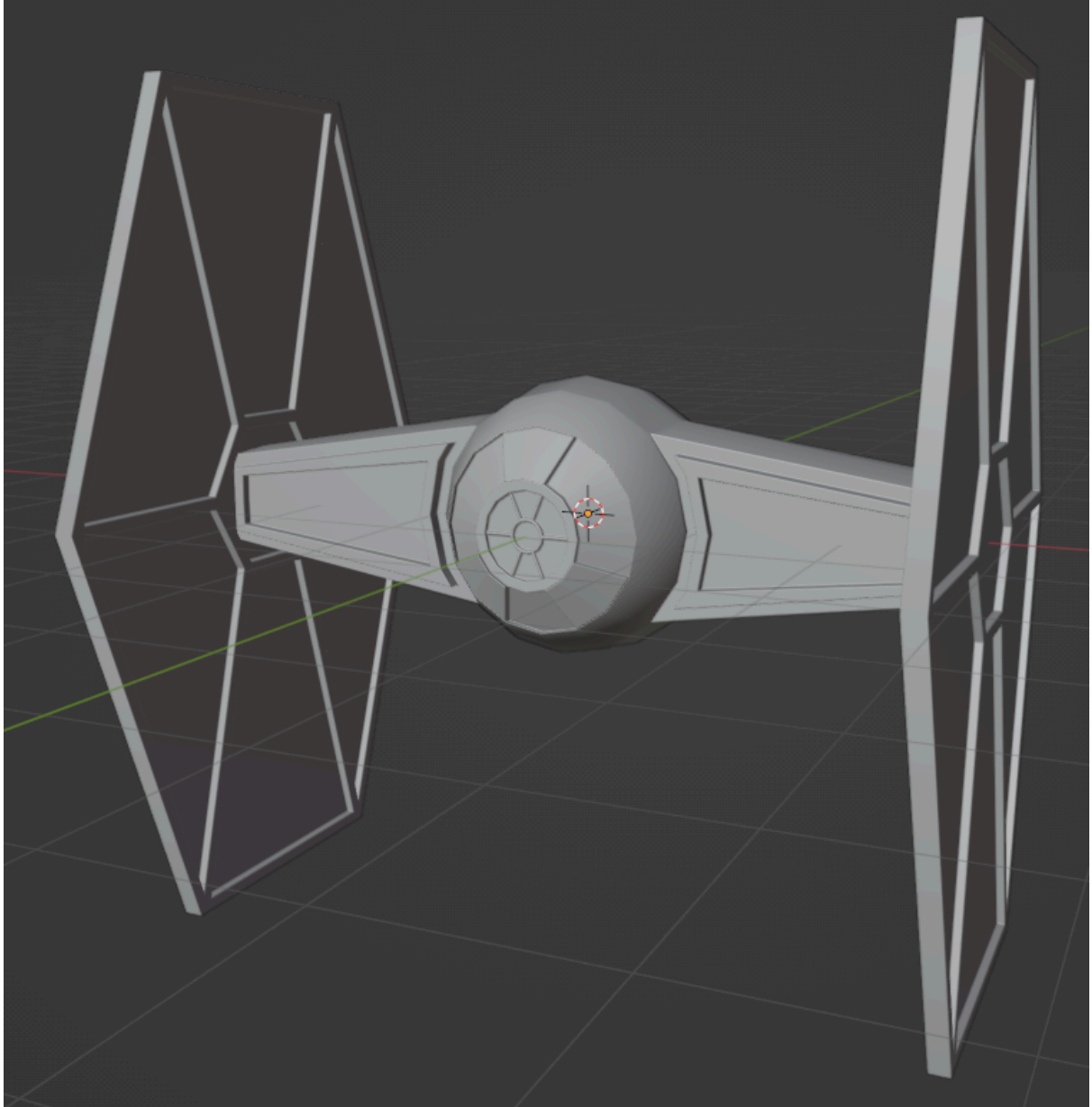
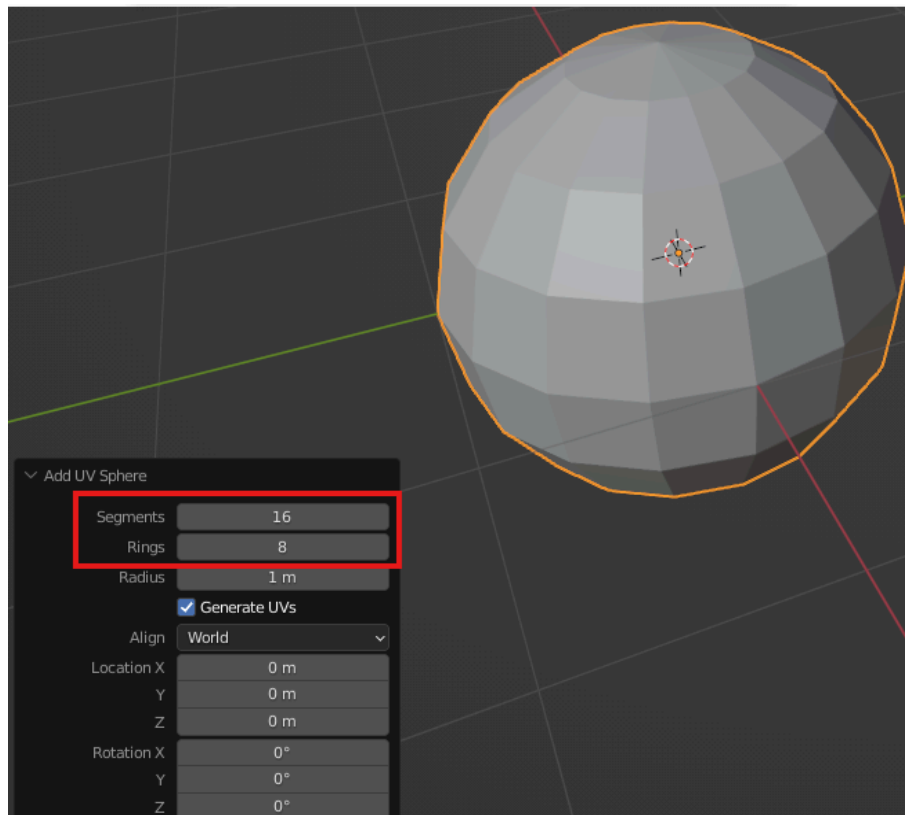


Tie Fighter tutorial

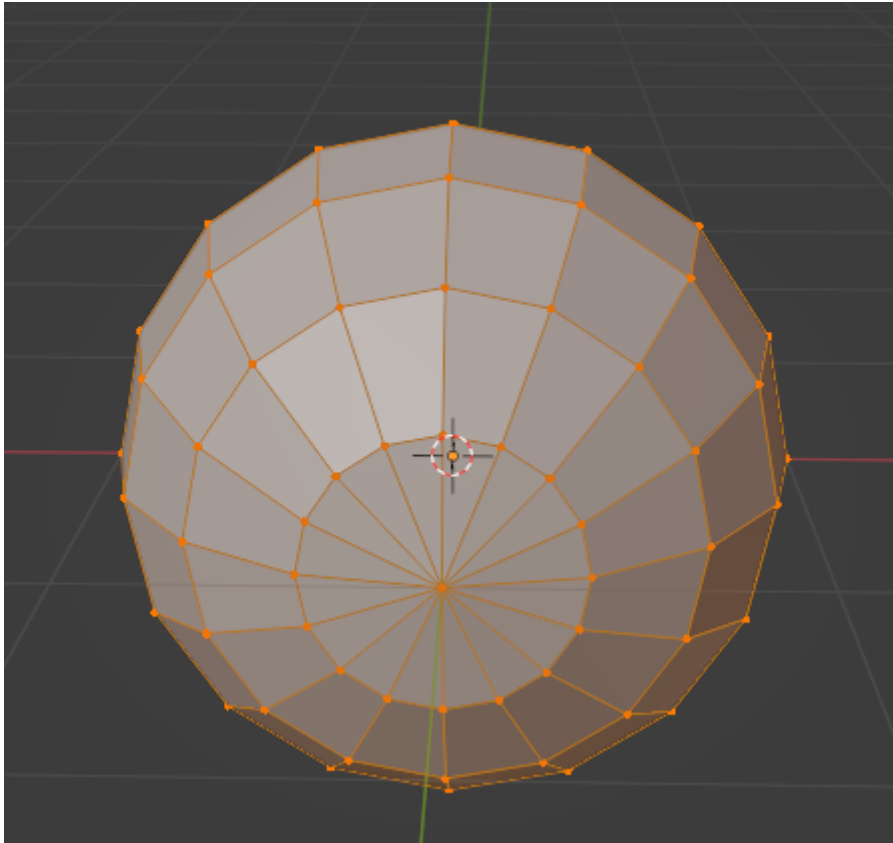
Important concepts covered: grid fill, mirror modifier



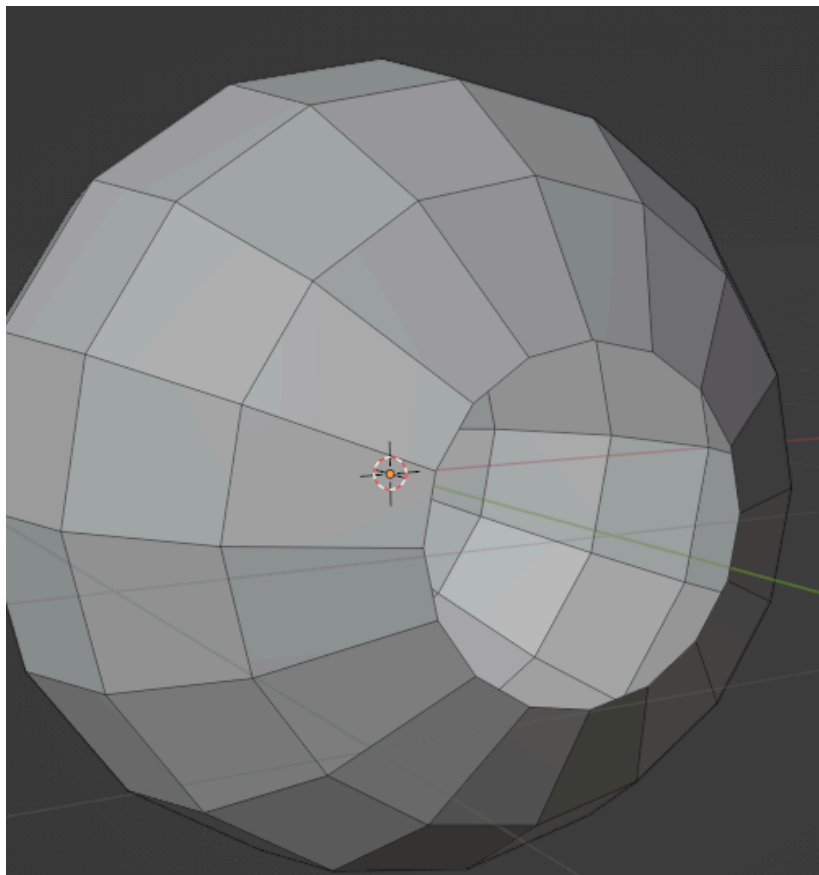
Let's start by deleting the default cube (DEL or X) and adding a UV sphere (SHIFT+A → MESH → UV Sphere). Change its properties to have 16 Segments and 8 Rings:



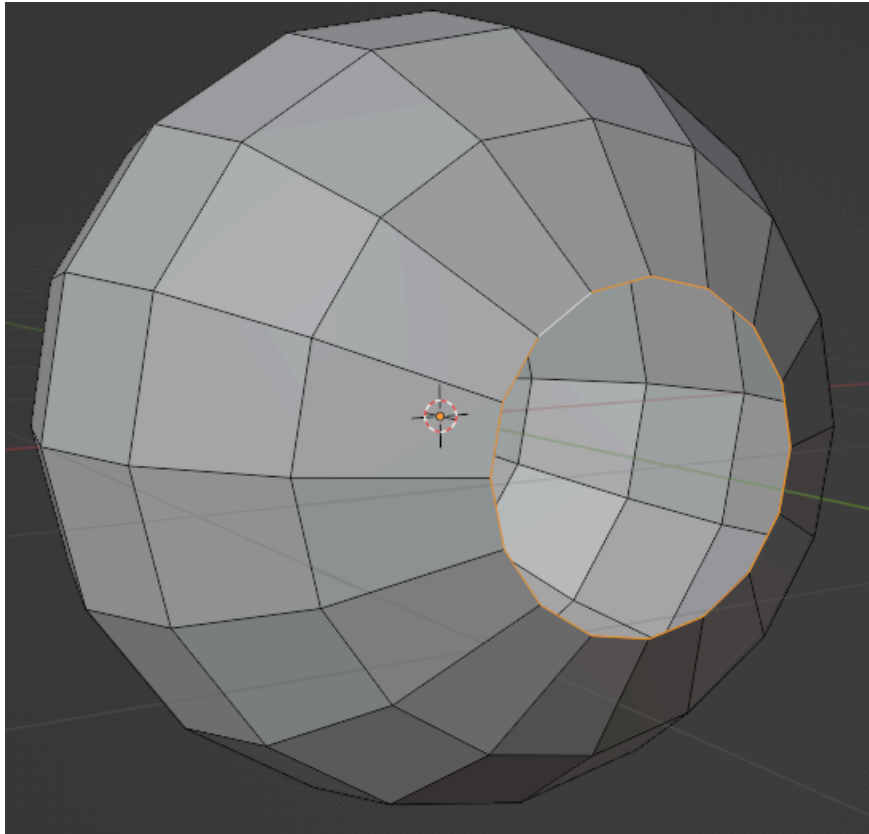
Then, rotate it along the X axis (R+X) by 90° and enter edit mode:



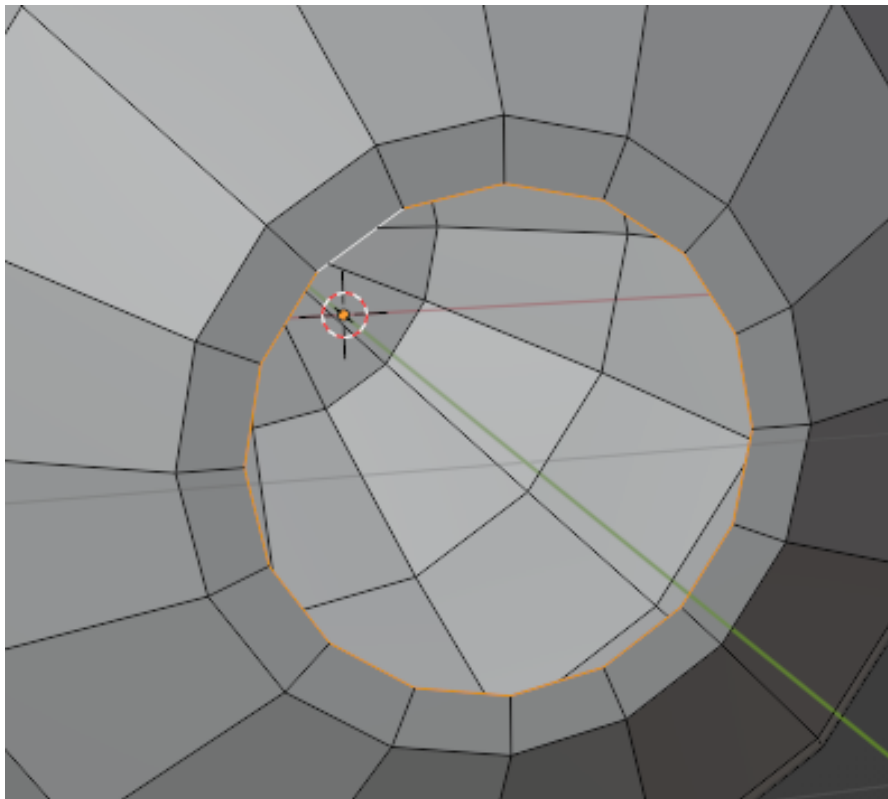
Enter Face selection mode, select the front faces and delete them (DEL or X):



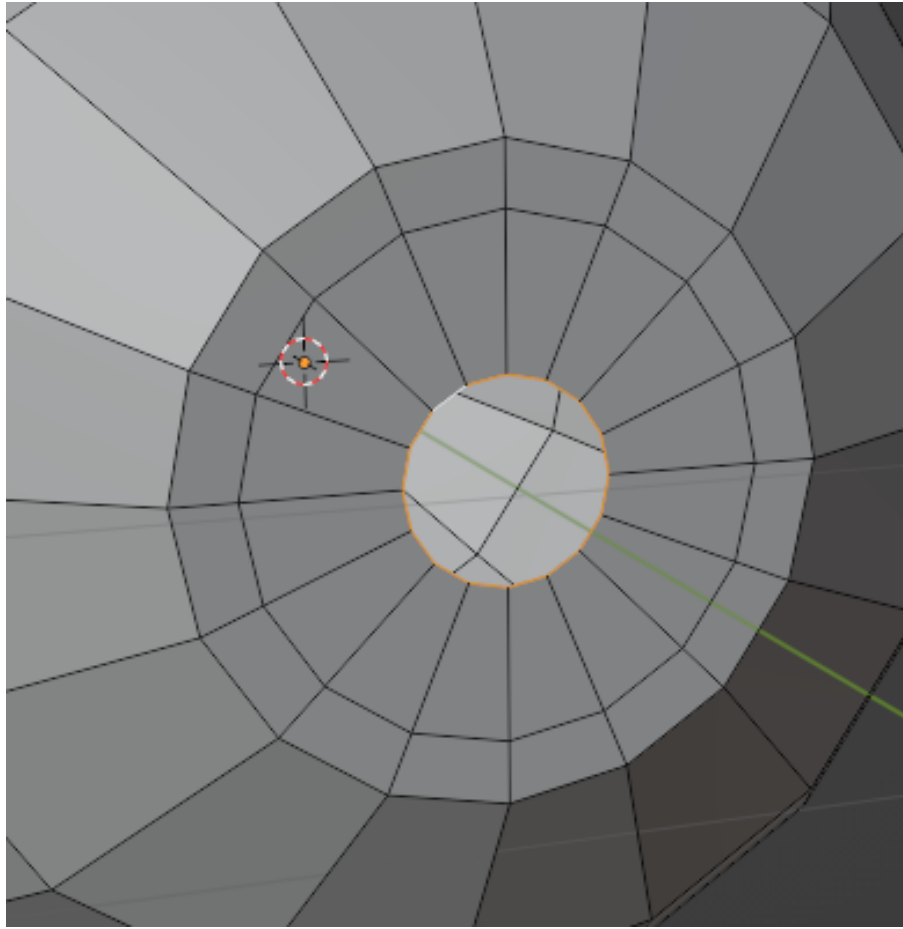
Now change to edge selection mode and select the edge loop created by holding ALT and clicking with the left mouse button on one of the edges of the loop:



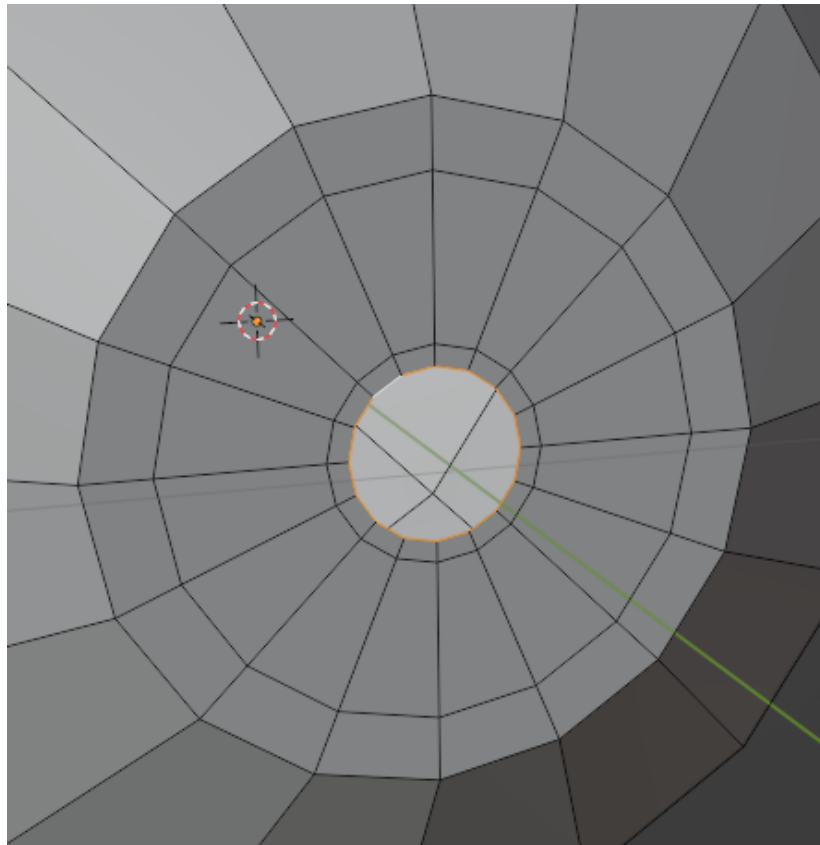
Now Extrude (E) and scale down (S) by 0.8 the selection:



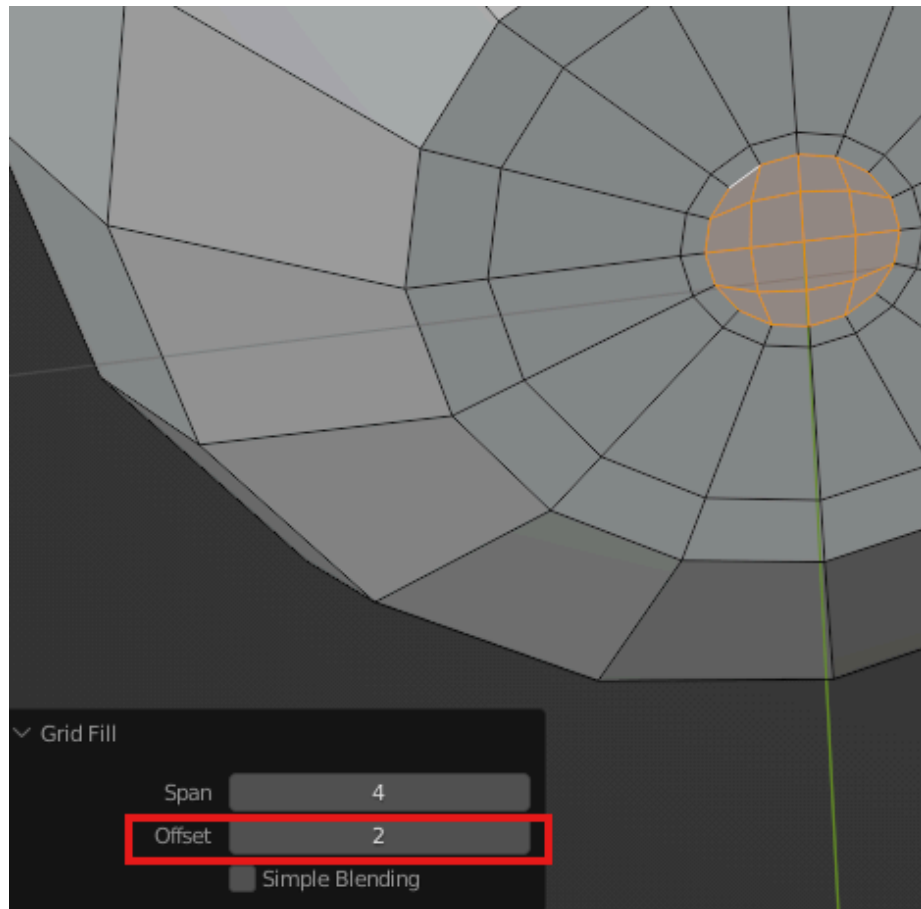
Repeat the process but now scaling down (S) by 0.4:



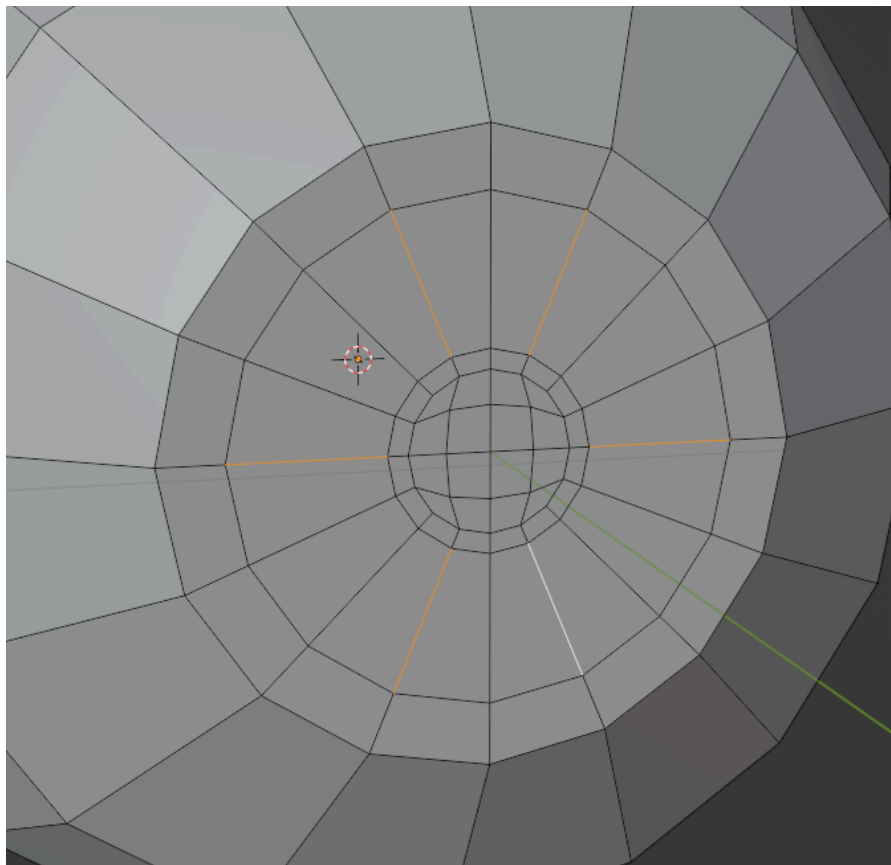
And again (S) by 0.8:



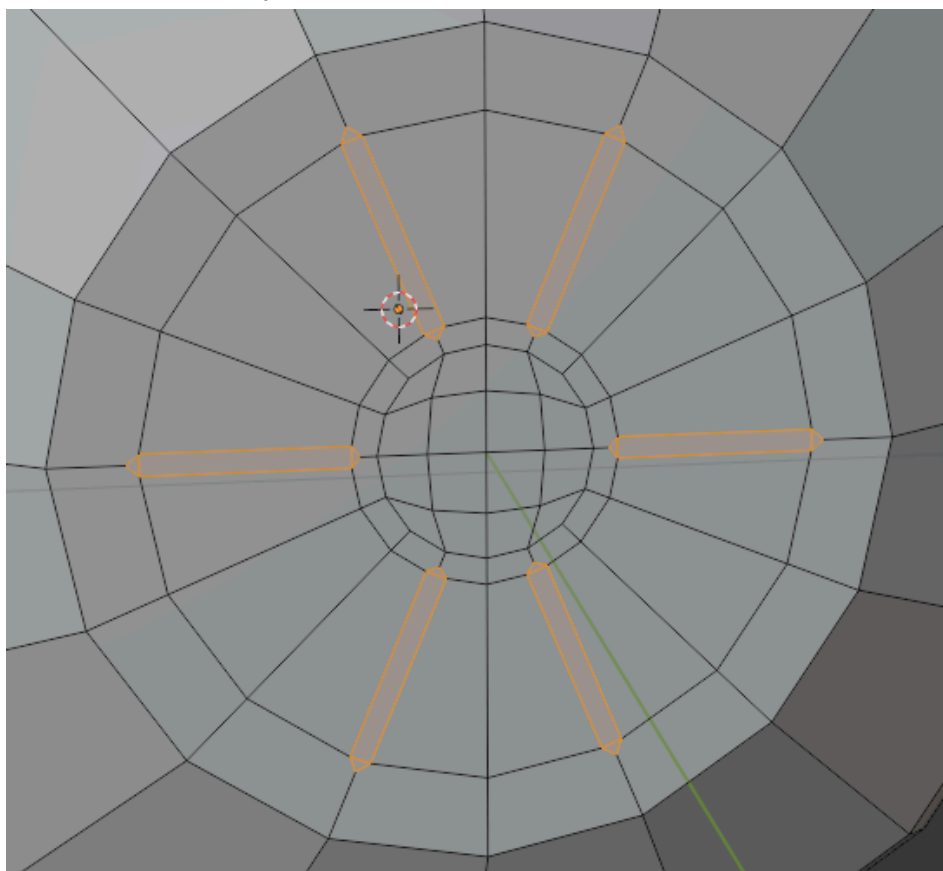
Now with this selection, press CTRL + F and select “Grid Fill”. Change the offset to be 2:



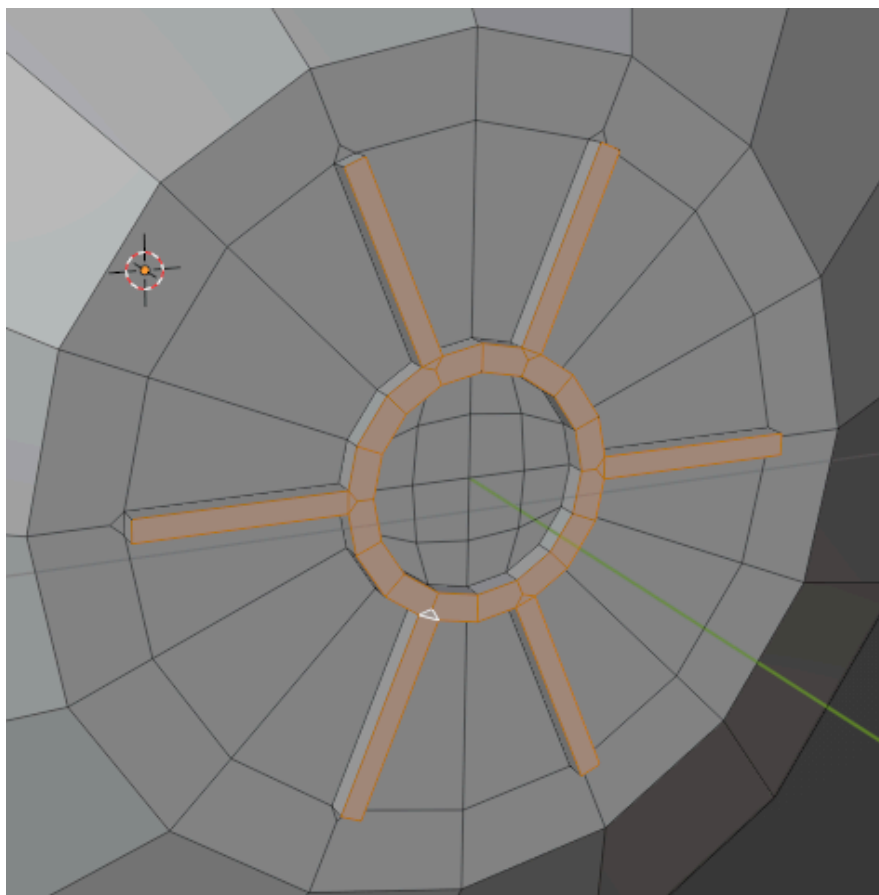
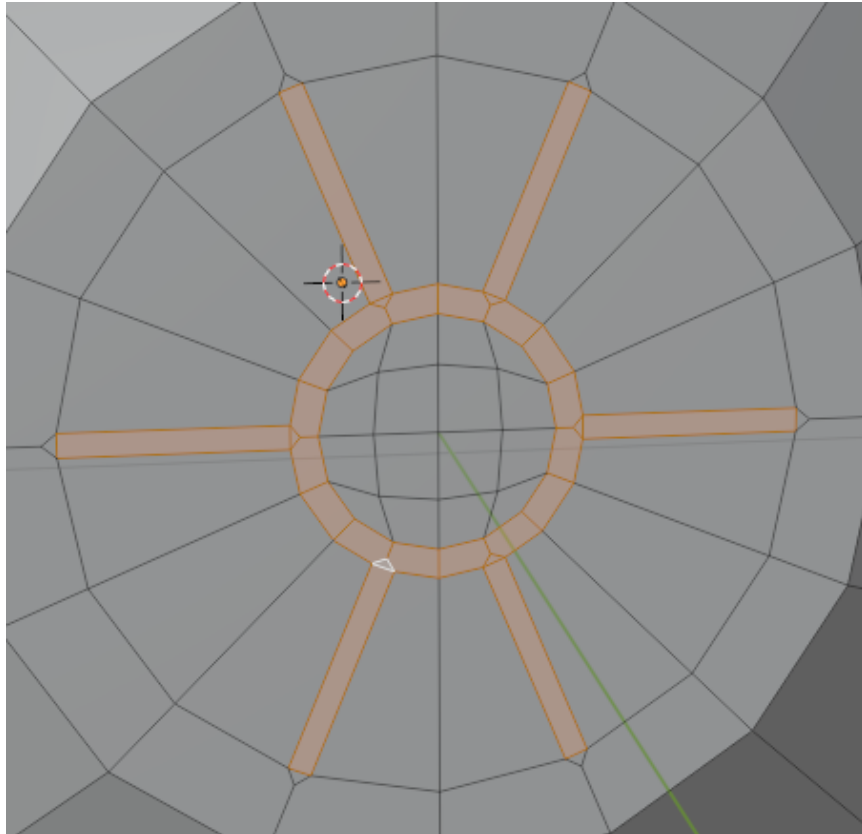
Now select the following edges:



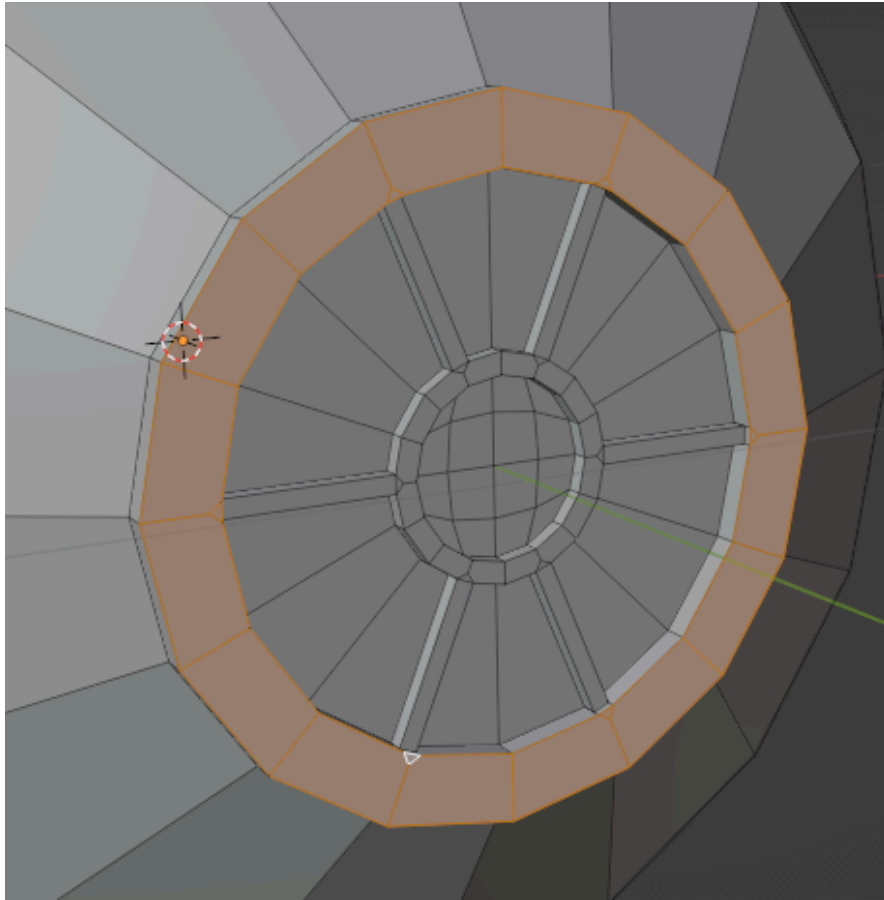
And bevel (CTRL + B) them by 0.01:



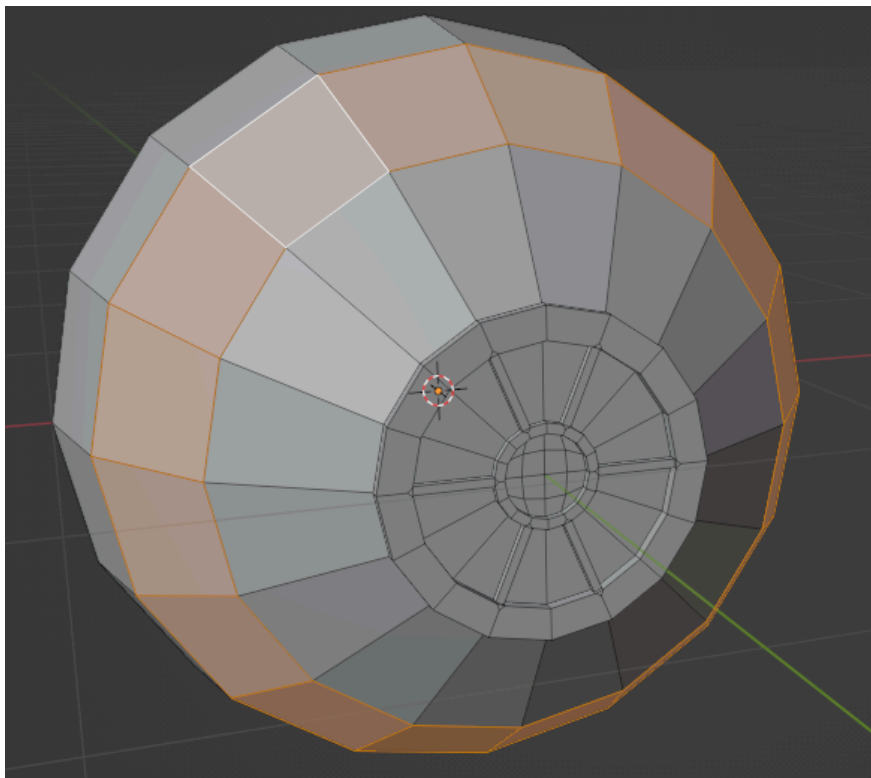
Now change to Face selection mode and select the following faces and extrude (E) them by 0.02 (pay attention to the little triangles created! you also have to select them!):

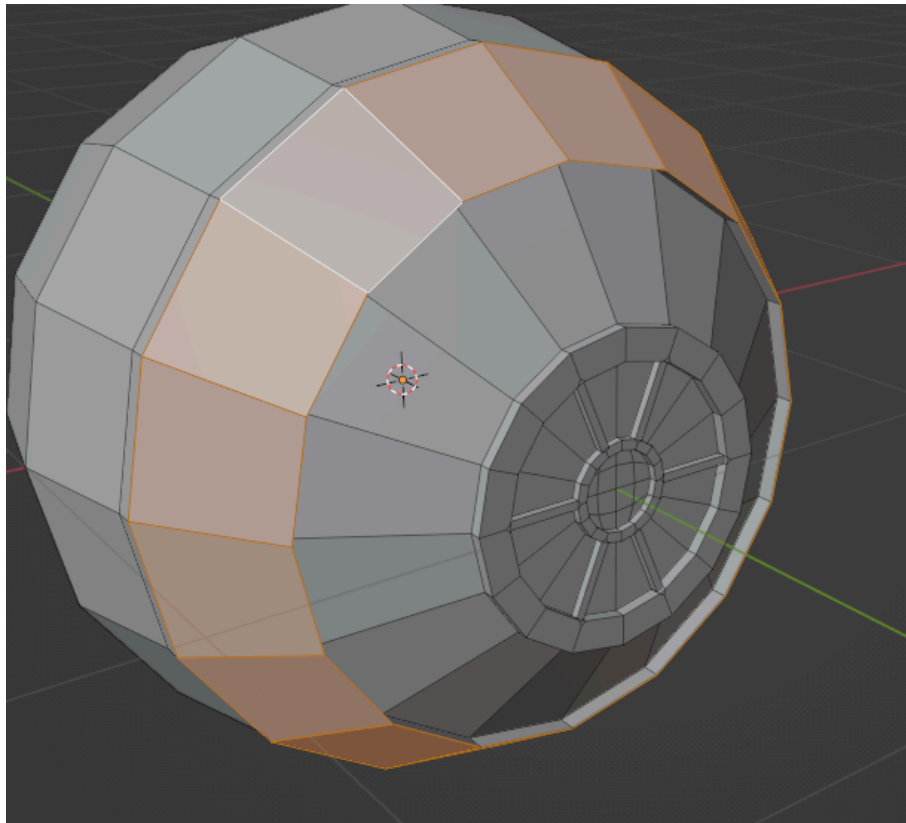


Now select the outer ring and extrude (E) by 0.03:

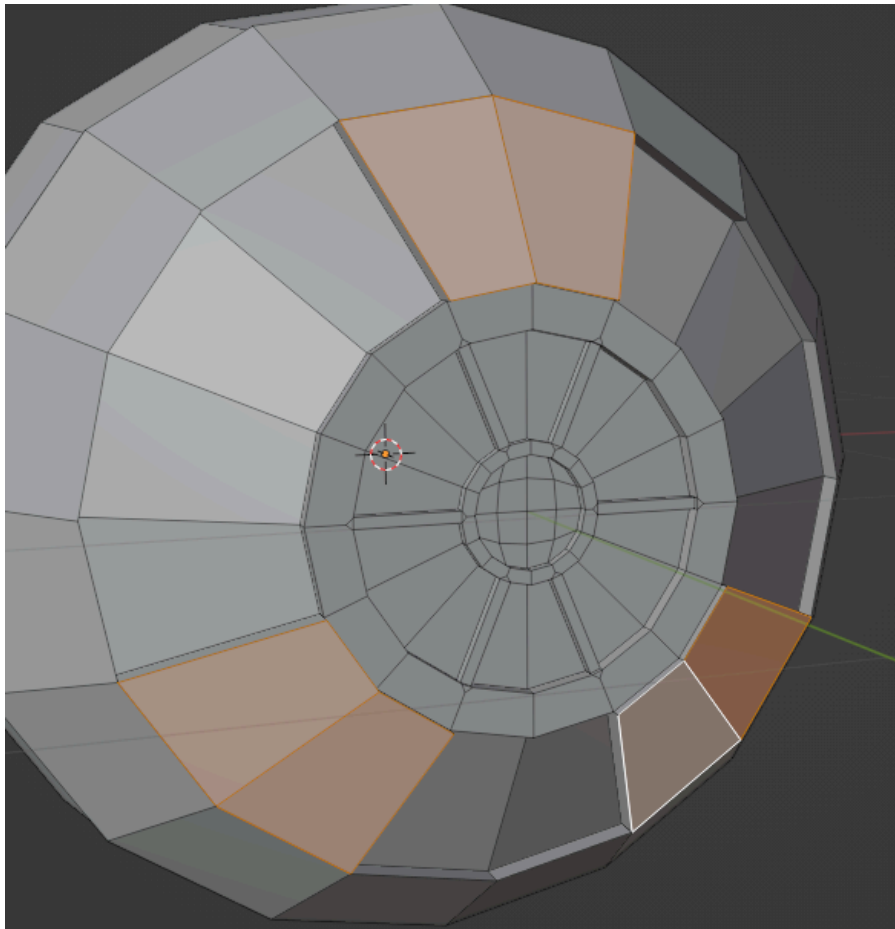


Now select the following face loop and extrude along Y axis (E+Y) by -0.05:

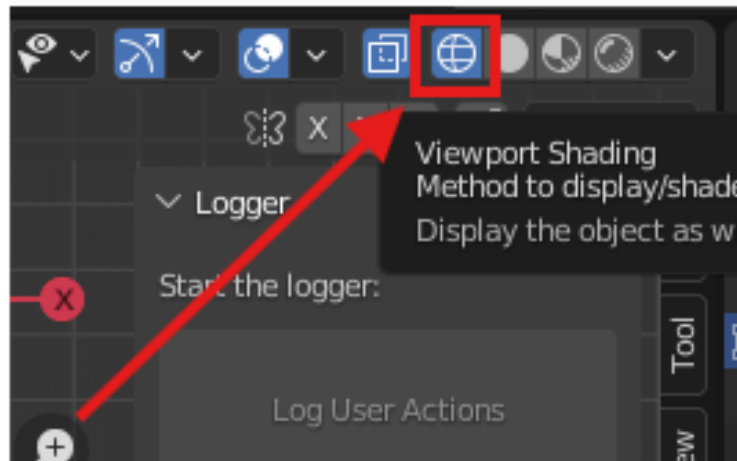




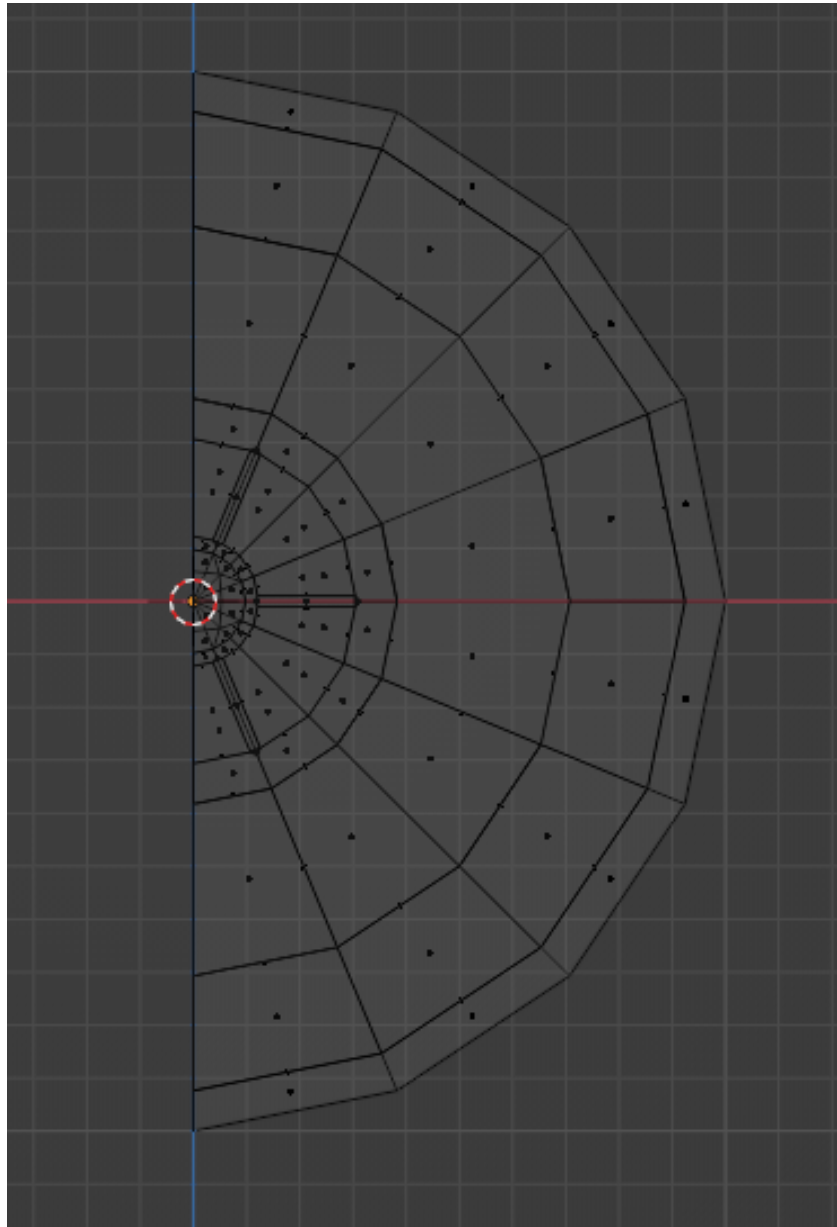
Next, select the following faces and extrude again along the Y axis (E+Y) by -0.05:



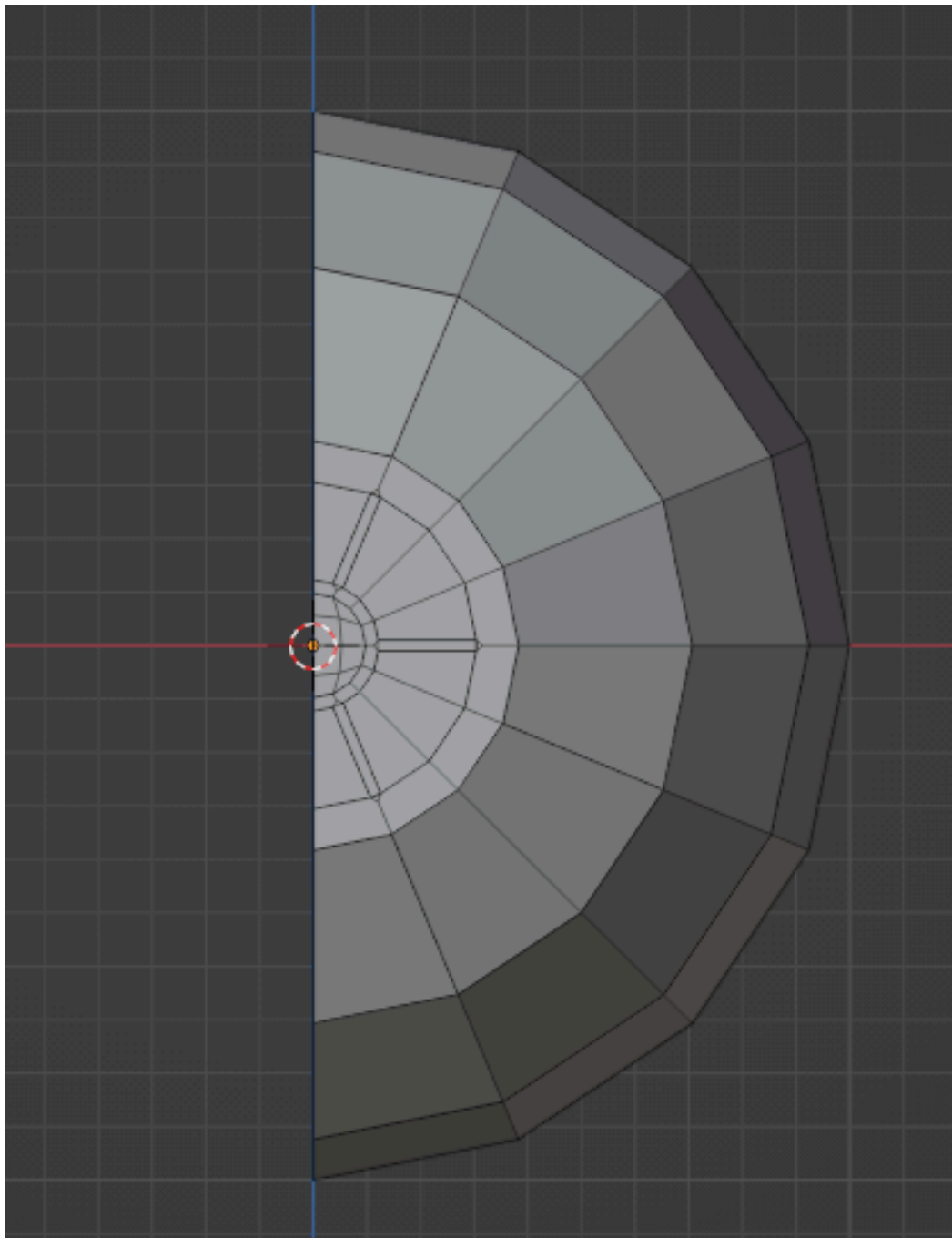
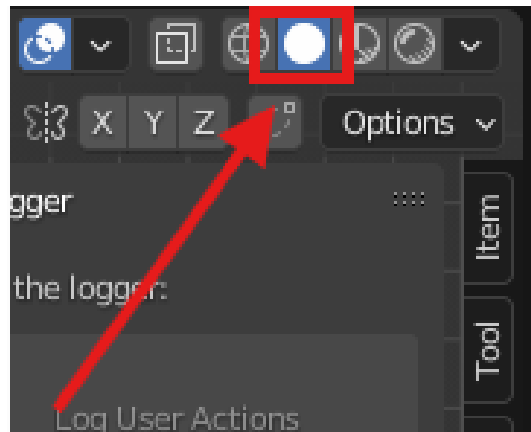
Now go to the front view by pressing NUMPAD 1 and change to wireframe mode by clicking on the button on the top right corner of the HUD:



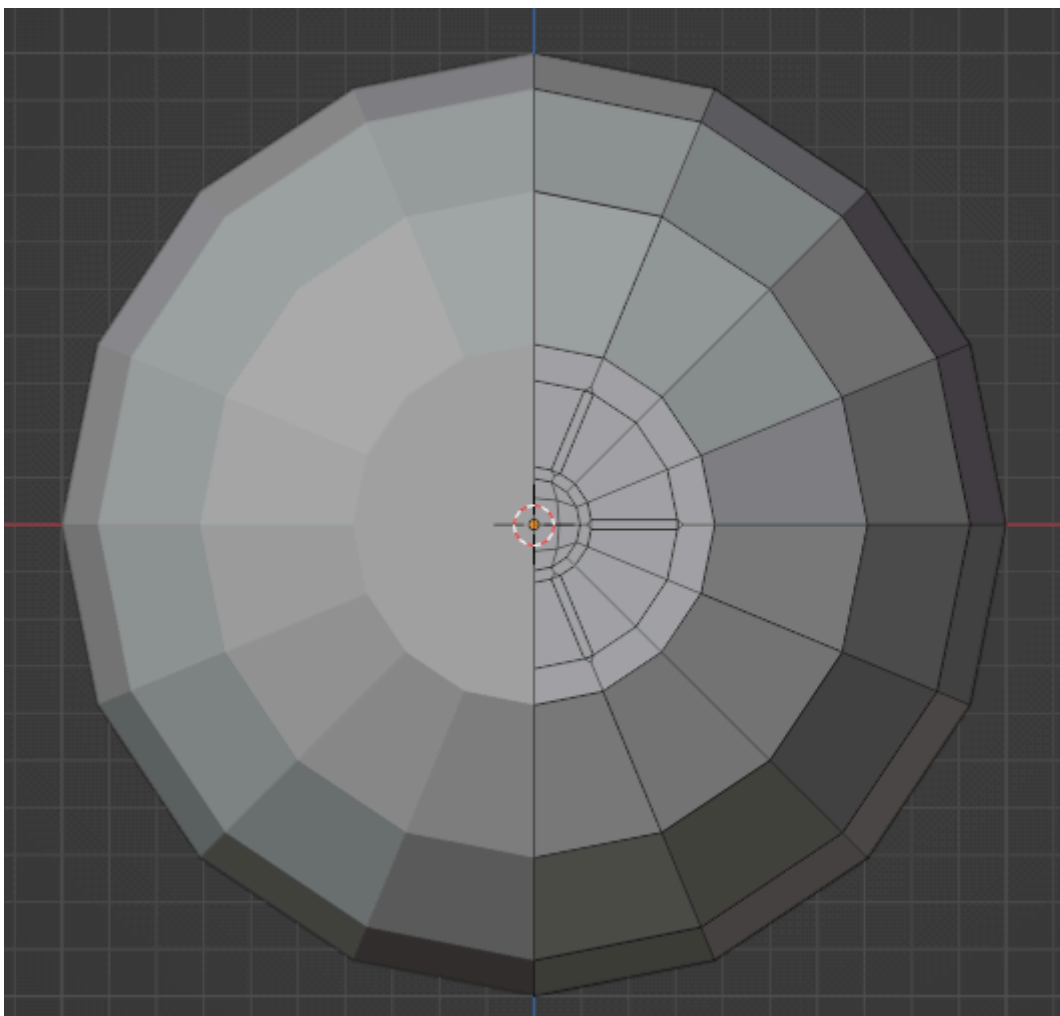
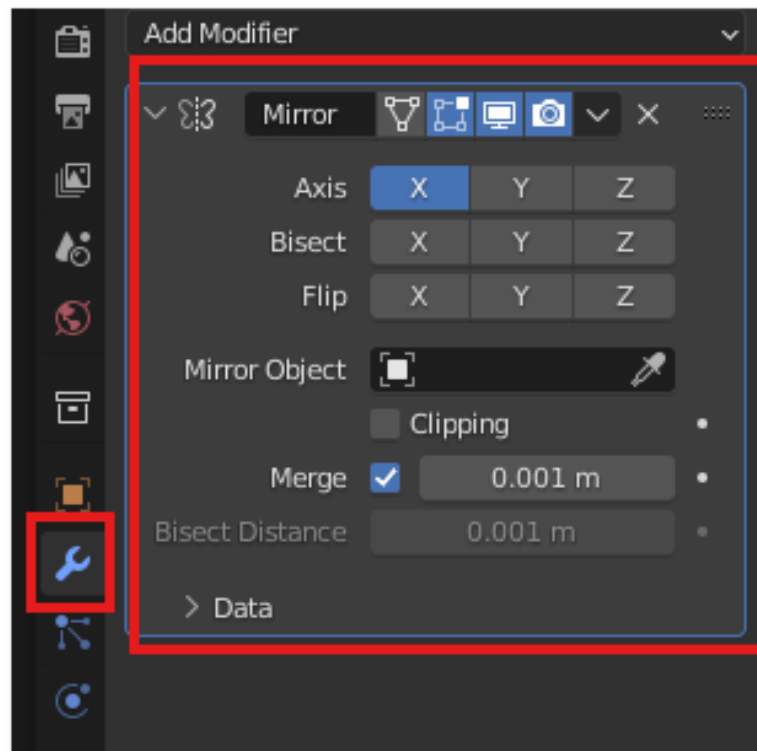
Click and drag the mouse to select the left half of the mesh and delete it (DEL or X):



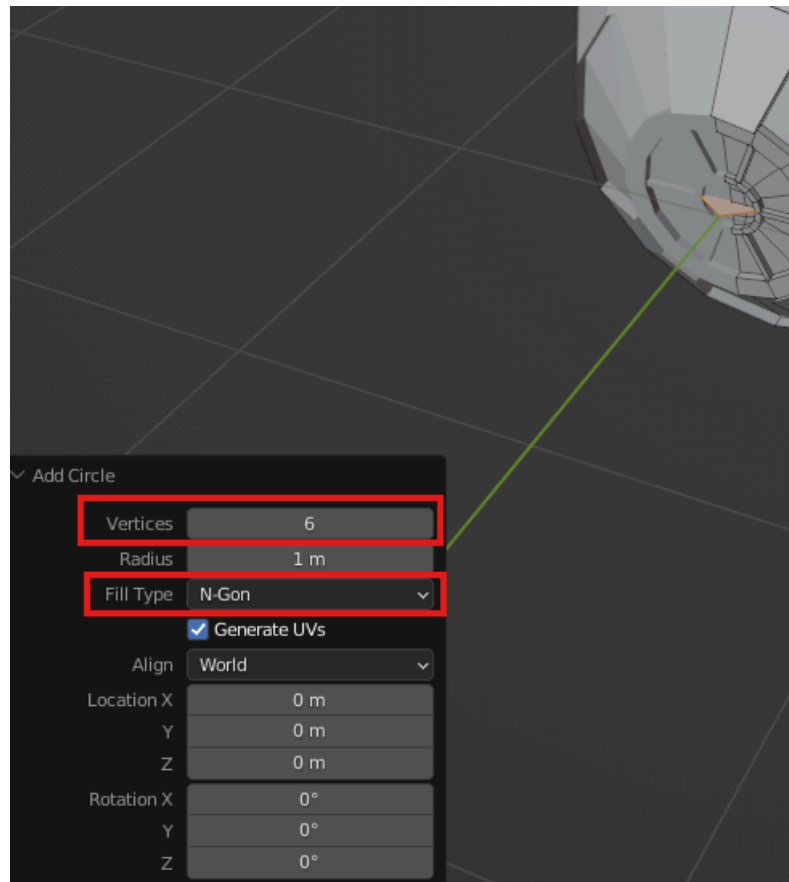
Now change back to the solid shading view:



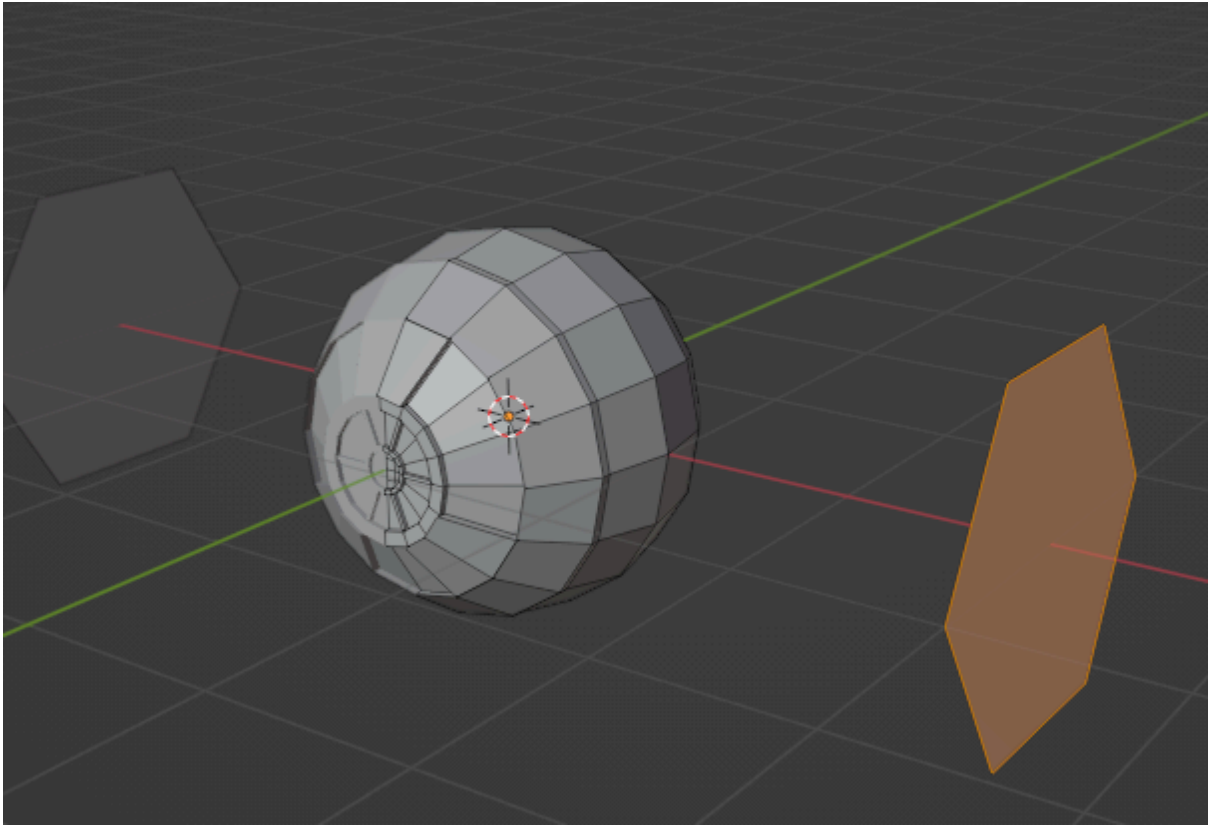
Next, add a mirror modifier:



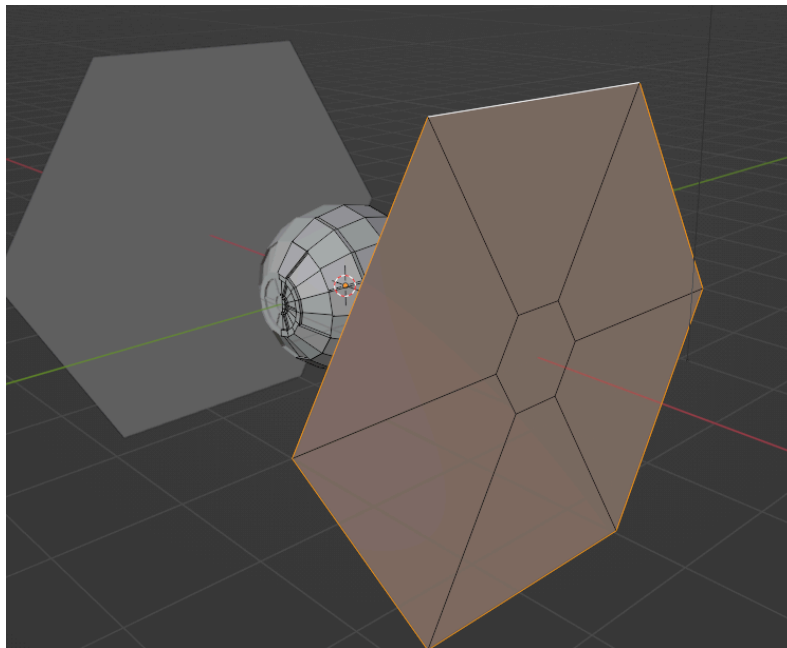
Now, make sure that the 3D cursor is at the center of the scene (you can press SHIFT + C to center it) and add a circle to the mesh (SHIFT+A → MESH → Circle) and select the number of the vertices to be 6 and the Fill Type to be N-Gon:



Move it along the X axis (G+X) by 3 and then rotate it along the Y axis (R+Y) by 90°:

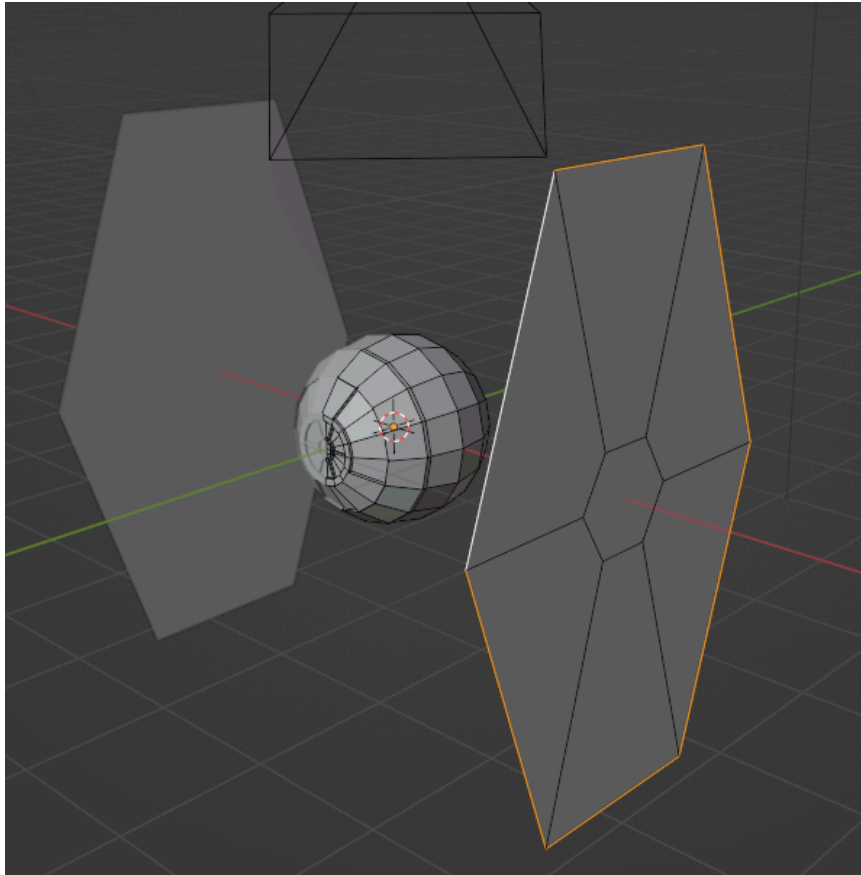


Scale it down (S) by 0.6 and change to edge selection mode. Next, with the hexagon selected, extrude (E) and scale it (S) up by 5:

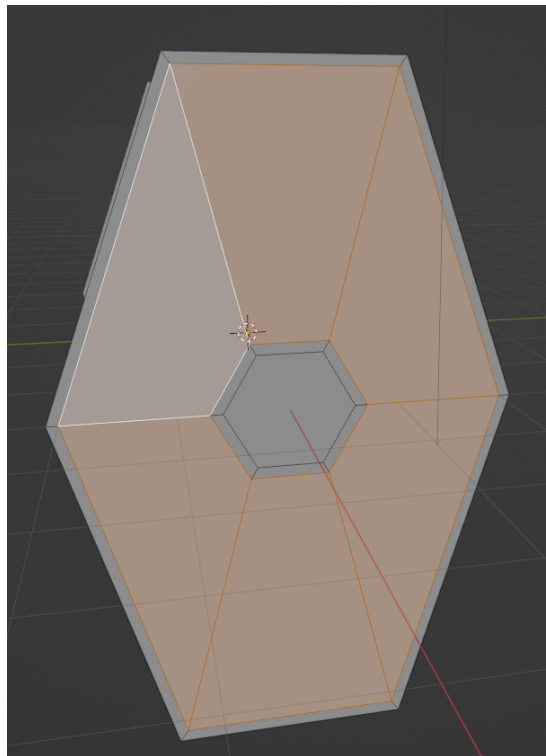


As you can see, we created an additional face that we don't want. Since the face is already selected, just press "X" to delete this face. In case you deselect it, just select it again making sure you are selecting the face that covers the entire hexagon and not only one of the parts of it.

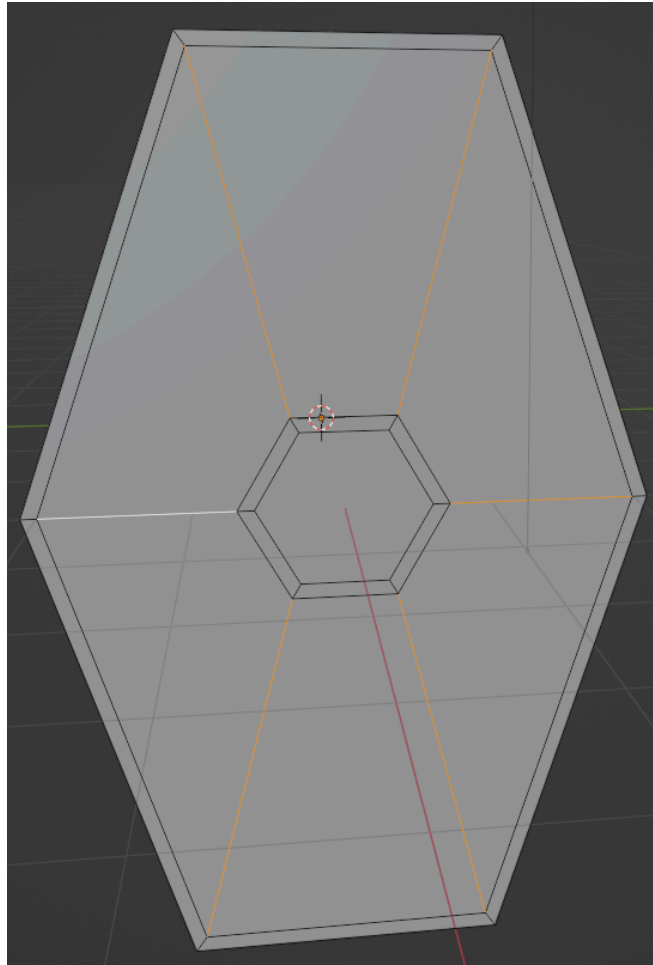
Now, select the outer edge loop and scale it along the Y axis (S+Y) by 0.7 and Z axis (S+Z) by 1.2:



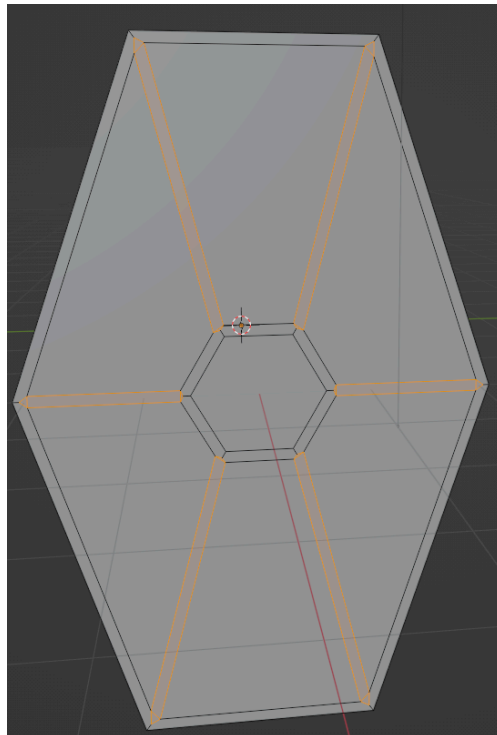
Now change to face selection mode and select the outer face loop and inset it by pressing “I” by 0.1:



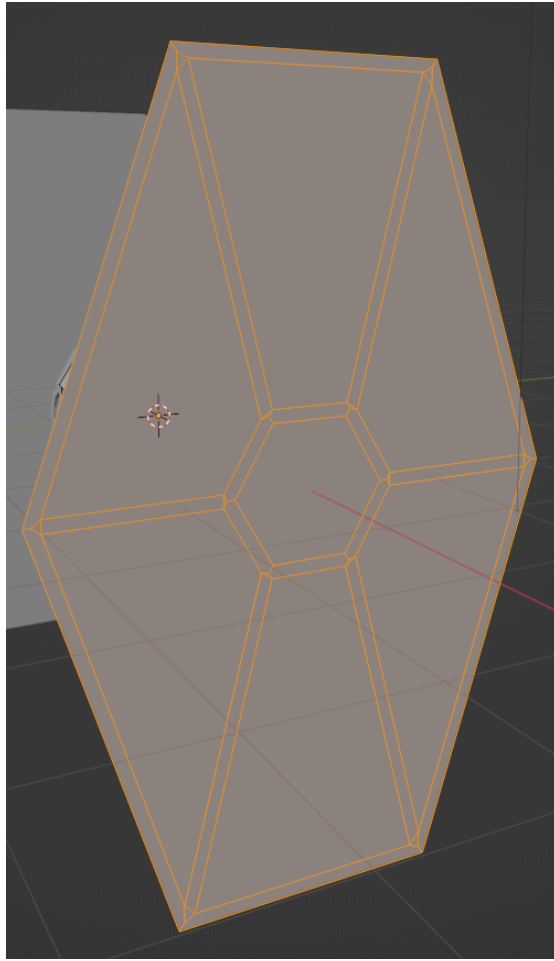
Now change back to edge selection mode and select the following faces:



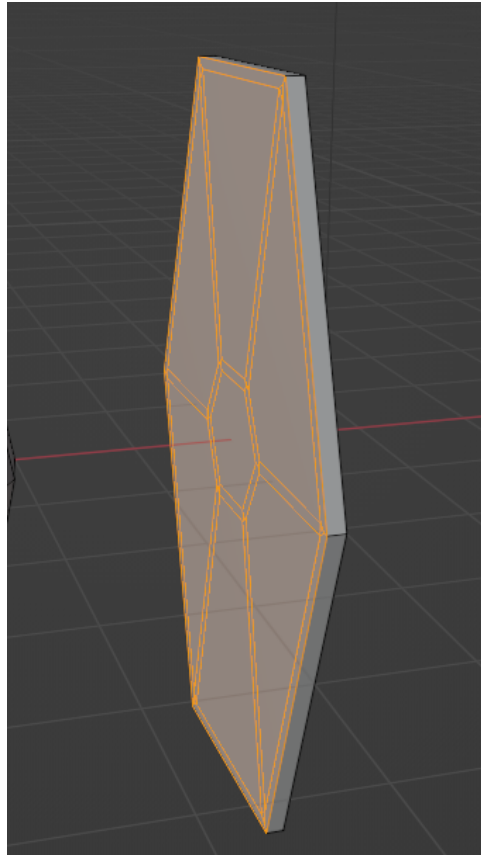
And bevel them (CTRL+B) by 0.05:



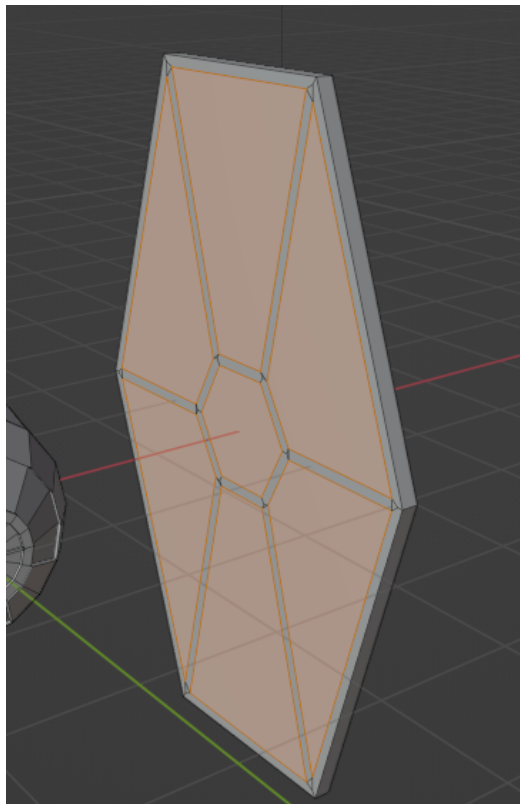
Now select the entire hexagon mesh. In order to do it faster, you can point the mouse to the mesh and press “L” to select all the linked mesh:

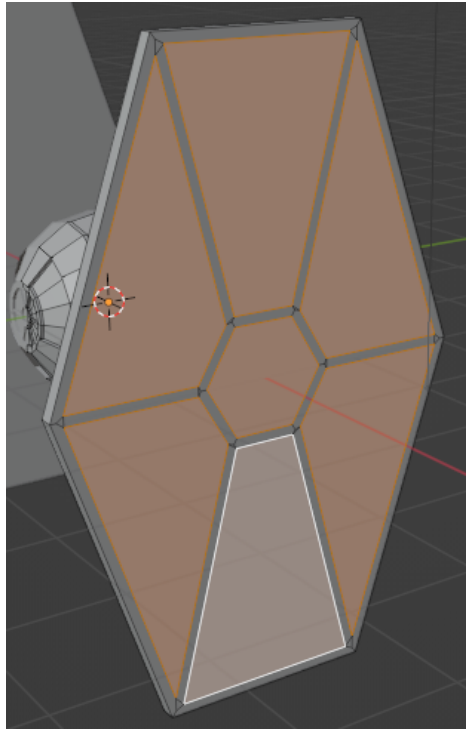


Make sure you are not selecting the central sphere!
Then extrude (E) it by 0.15:

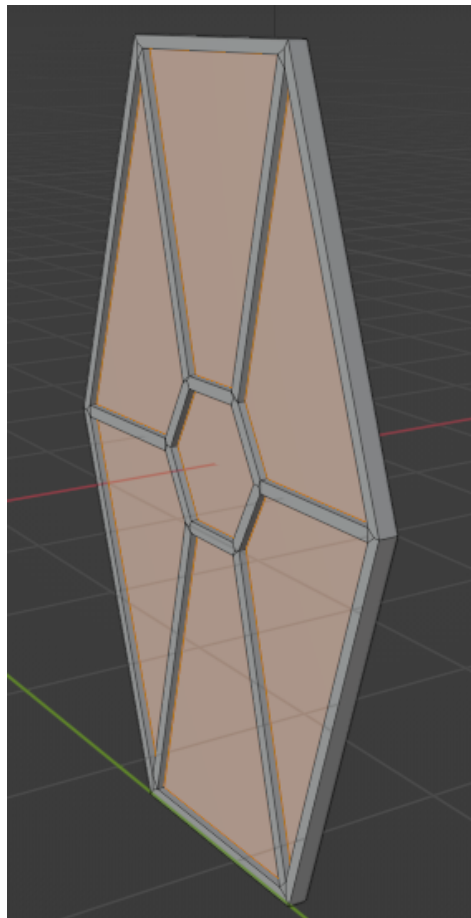


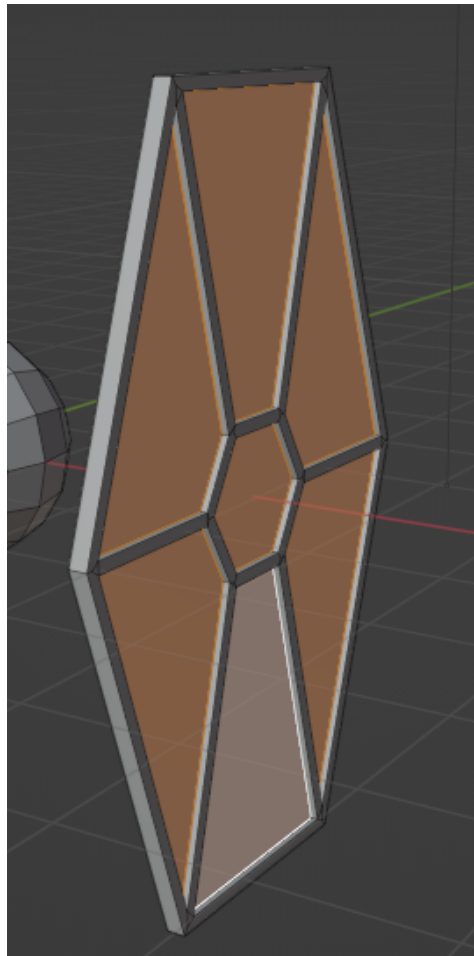
Now change to face selection mode and make the following selection **on both sides**:



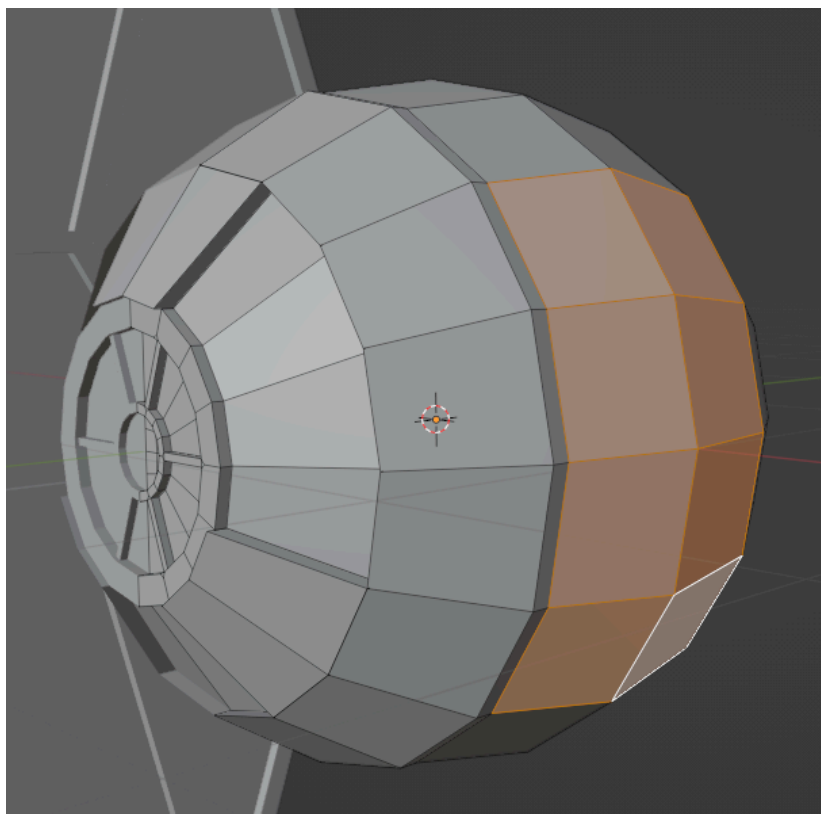


Extrude (E) and scale them down along the X axis (S+X) by 0.2:

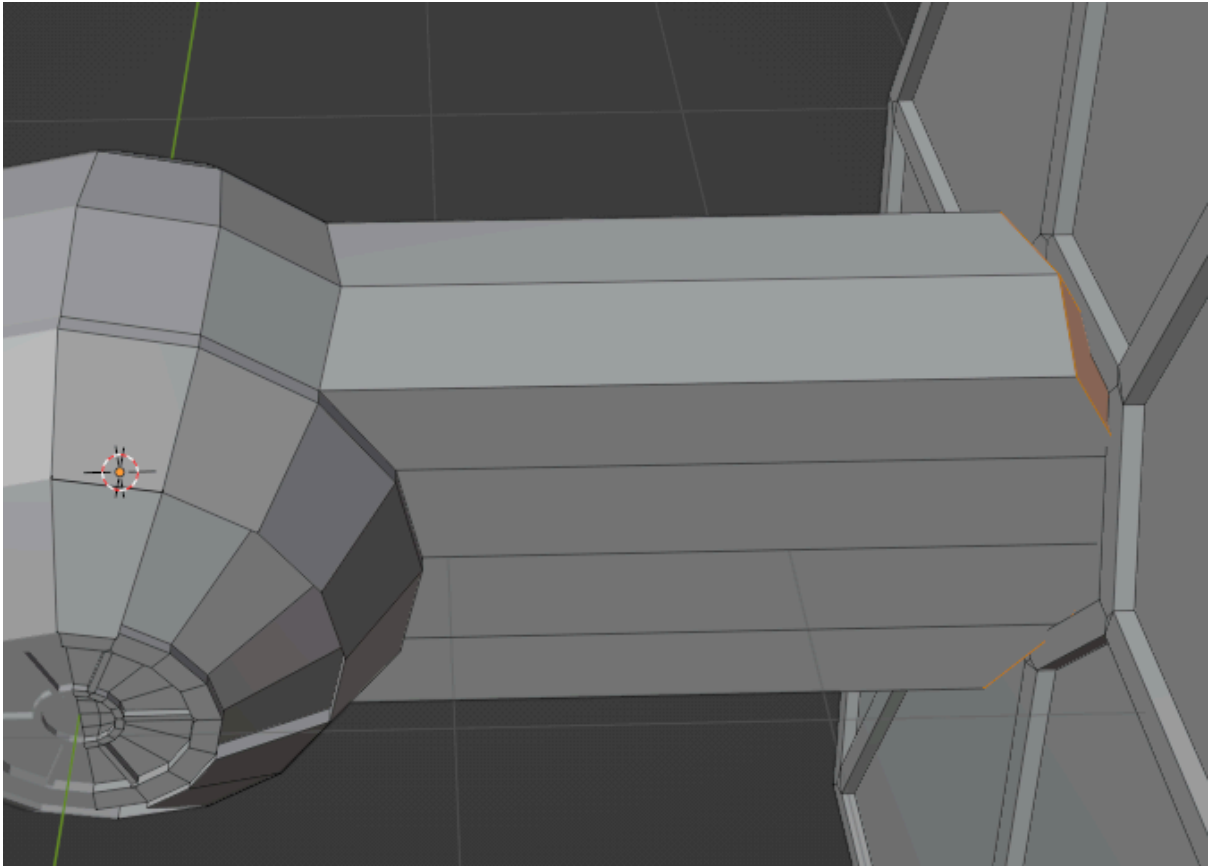




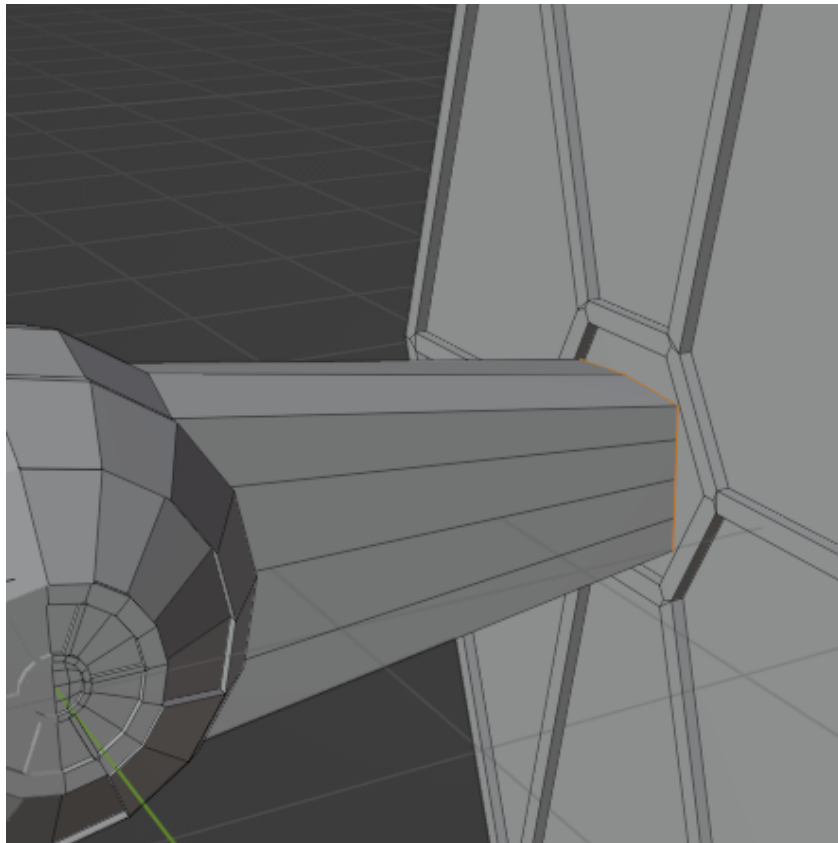
Now select the 8 middle faces of the sphere like the following:



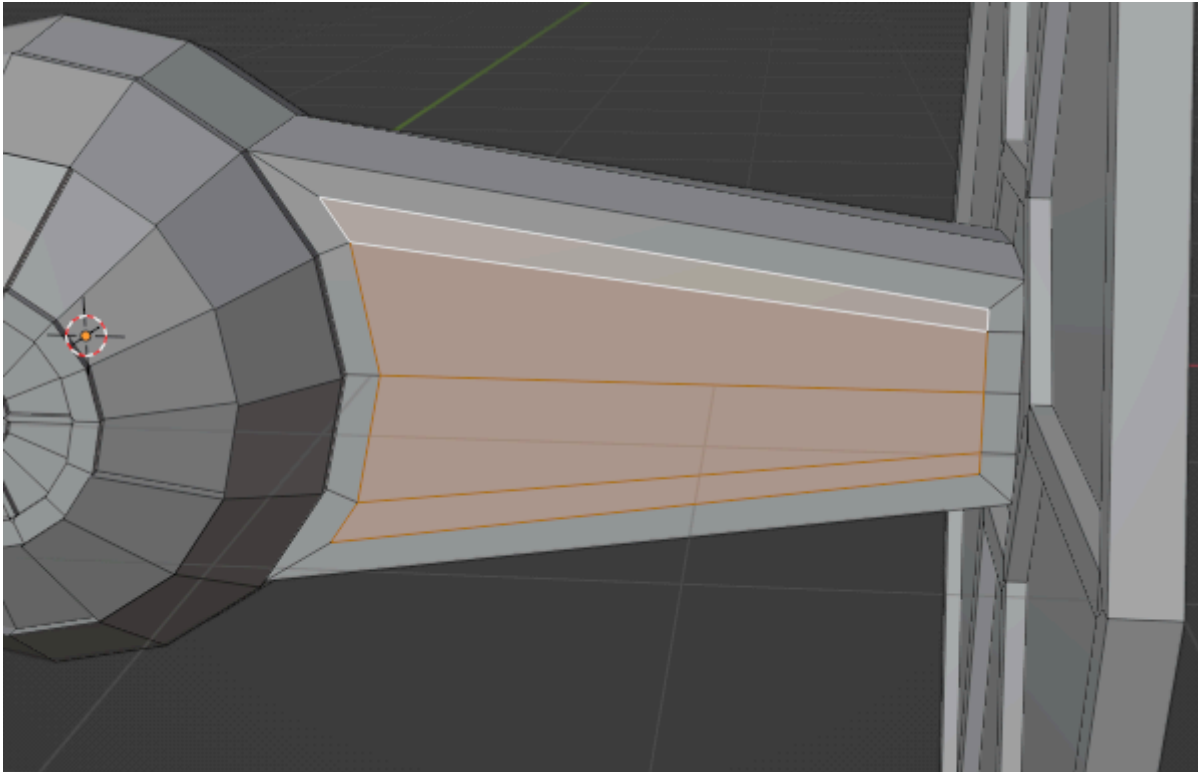
Extrude (E) the selection by 2.05:



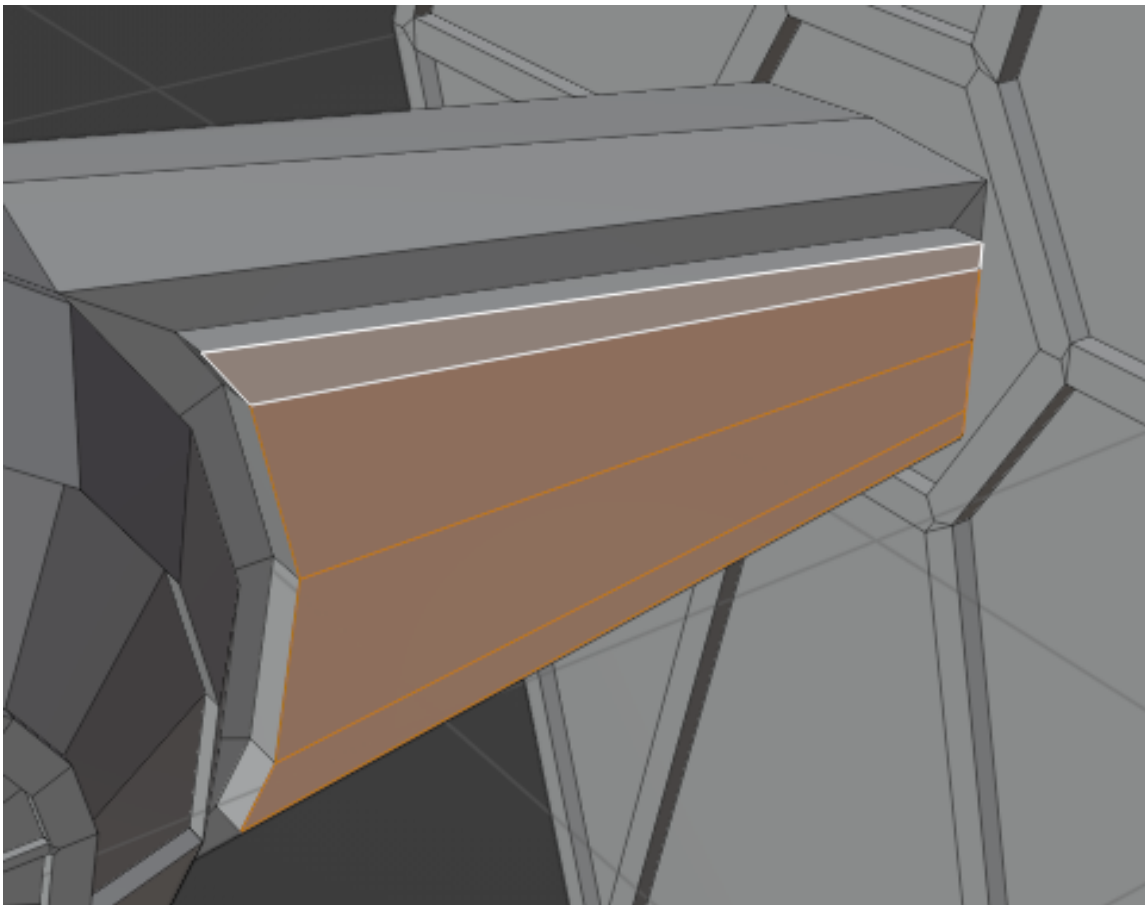
Next, with the same selection, scale it along the X axis (S+X) by 0 to flat it and then scale it along the Z axis (S+Z) by 0.5:



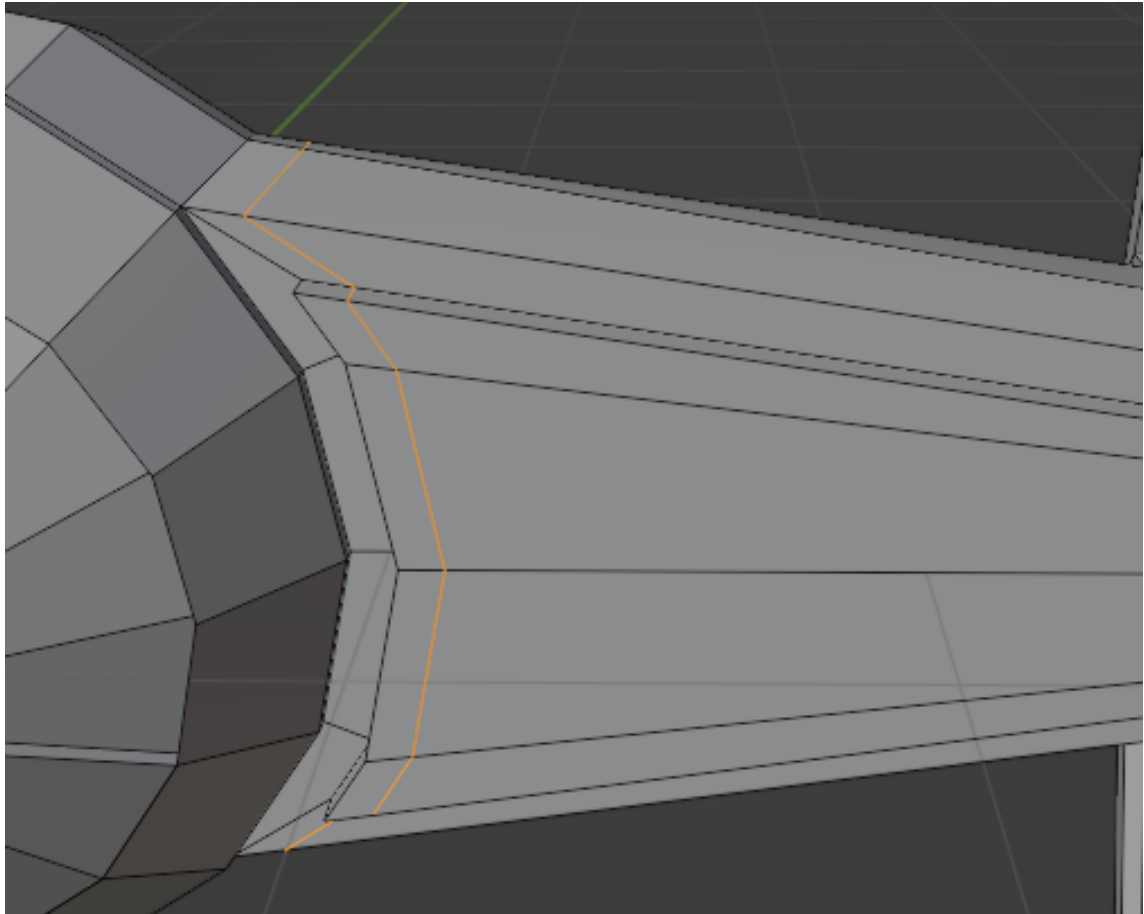
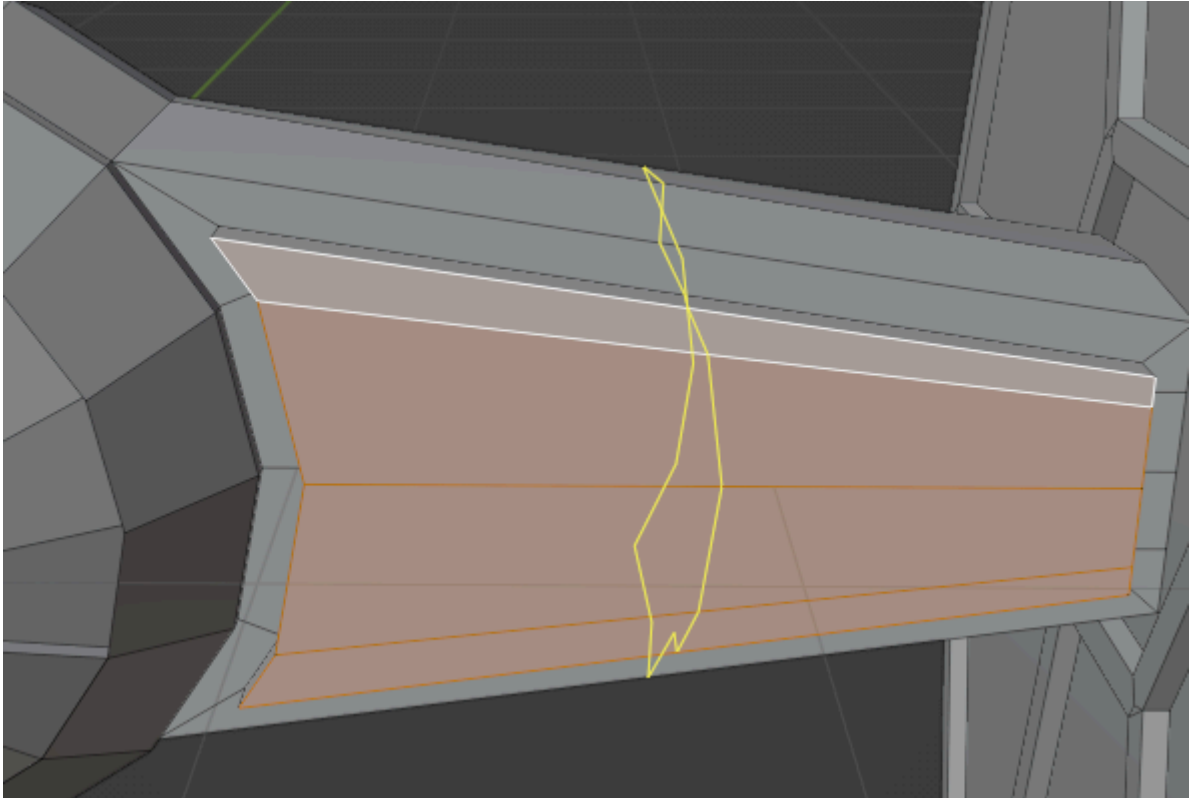
Now select the front faces created and inset (I) them by 0.1:



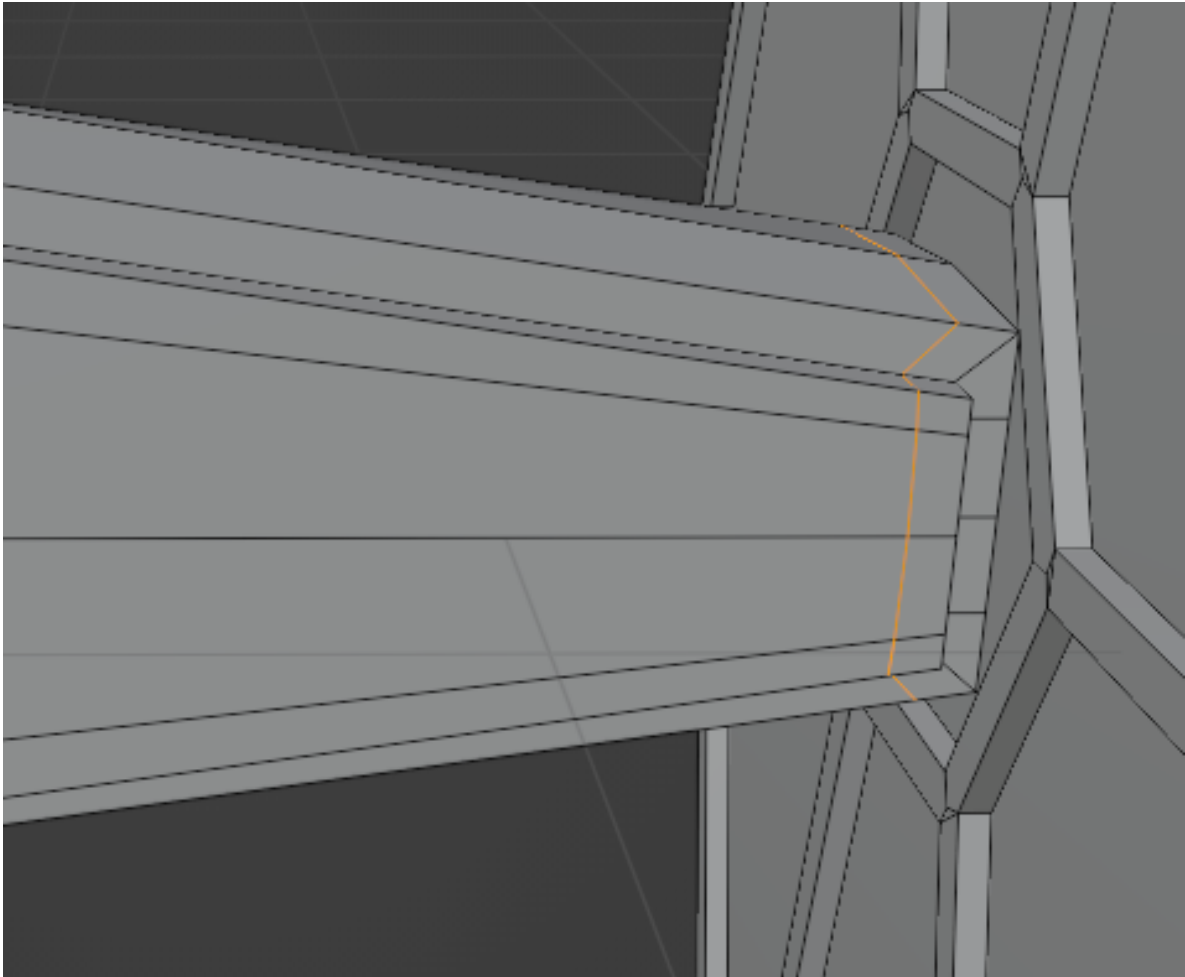
Extrude (E) it by 0.1:



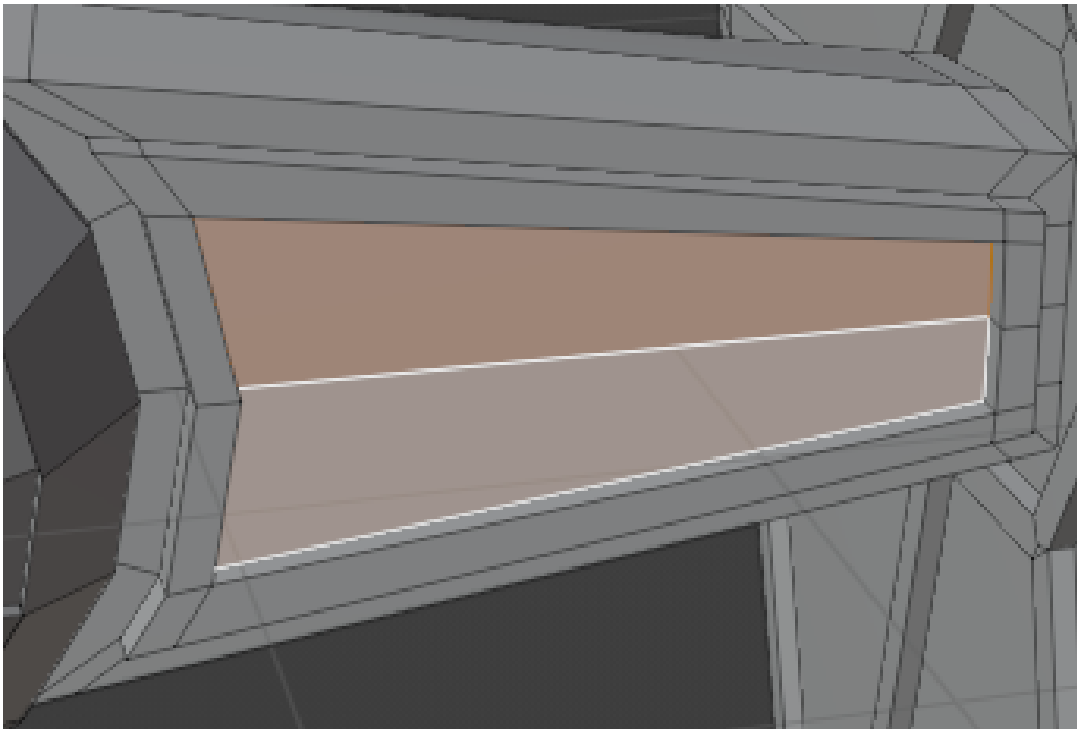
Now add a loop cut (CTRL + R) and move it by 0.9. To do so, press CTRL + R then select where you want to cut. Then, click once and type "0.9" so it will slide. Then click again to confirm:



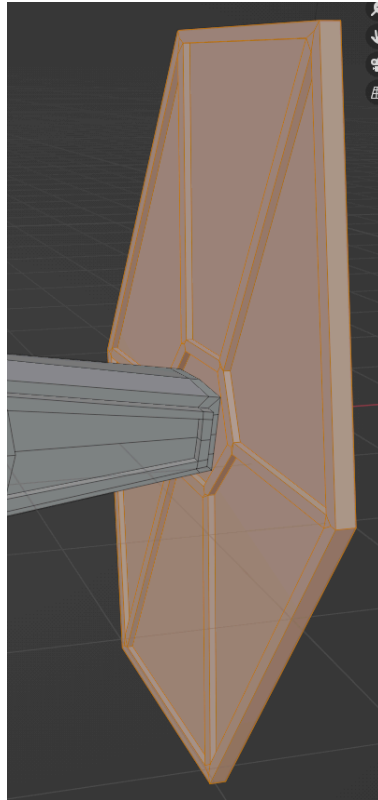
Repeat it again but now for the other side, sliding it by -0.9:



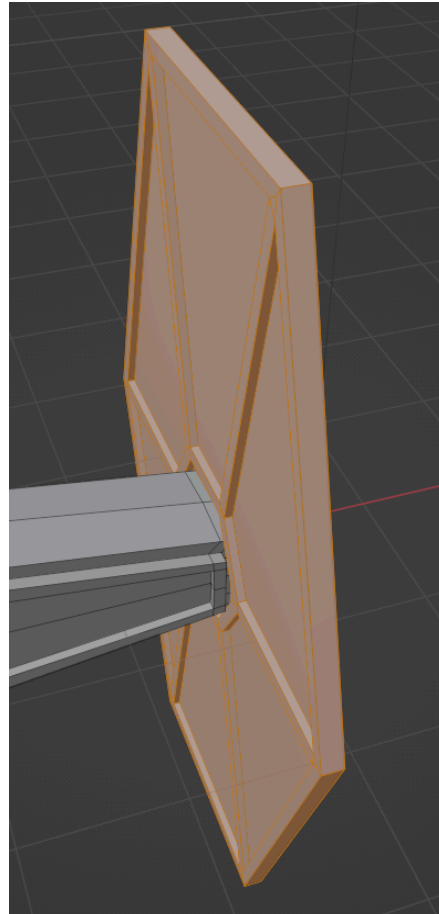
Change to face selection mode and select the middle faces and extrude (E) them by -0.07:



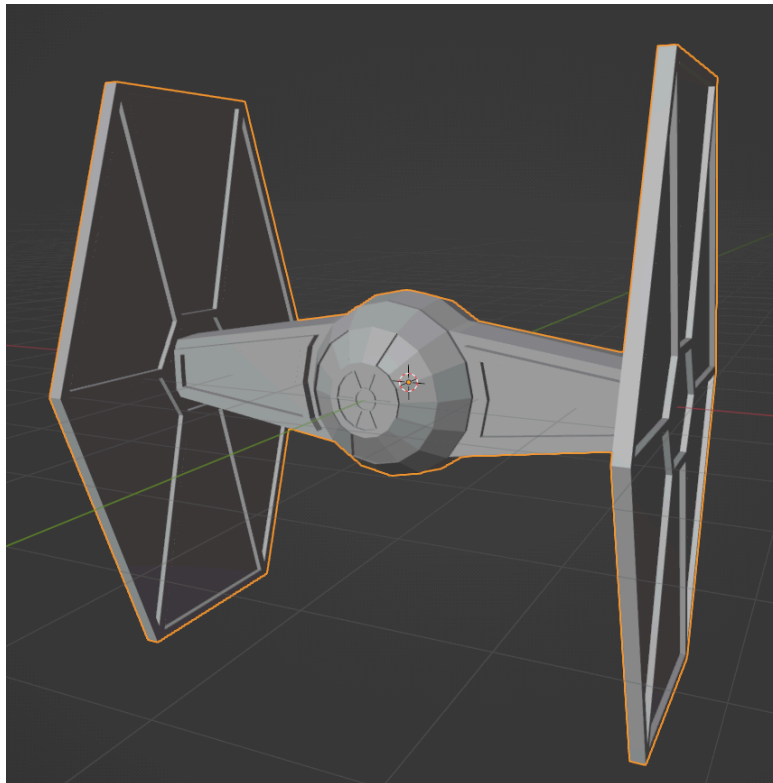
Now deselect everything by pressing “A” twice and select the hexagon again by pointing the mouse on top of it and pressing “L”:



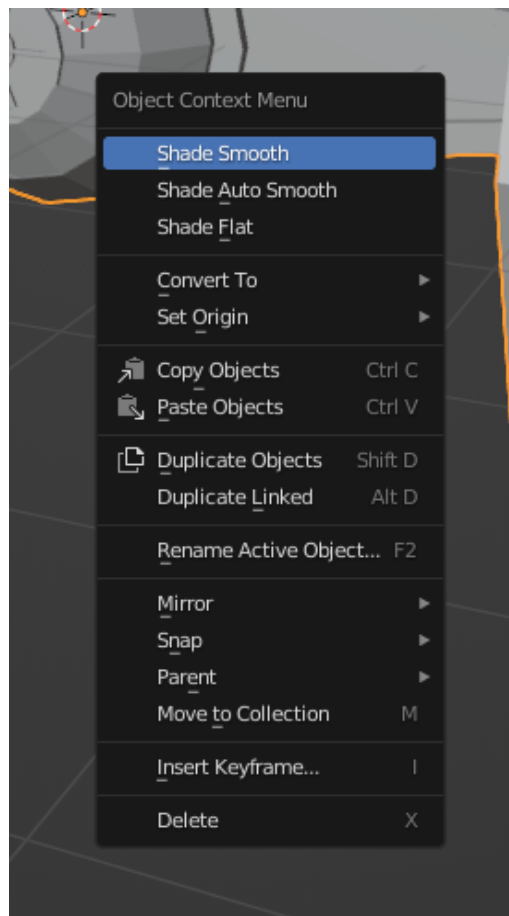
Move it along the X axis (G+X) by -0.08:



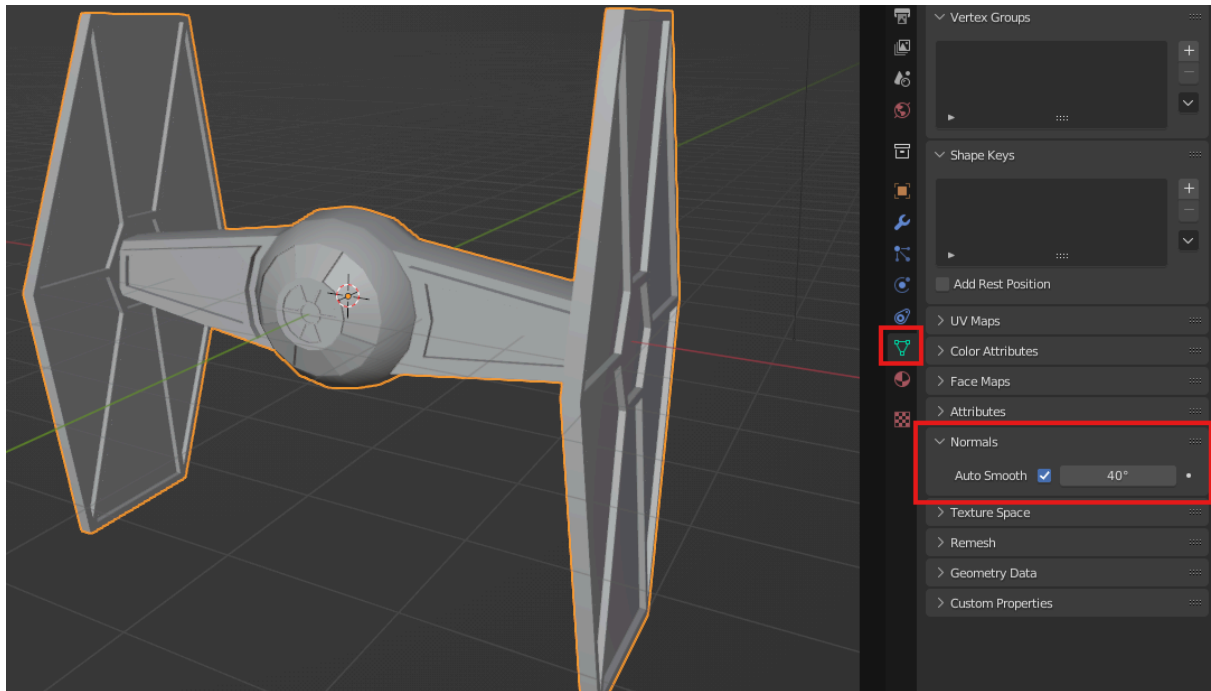
Now change to object mode:



Now, click with the right mouse button and select “Shade Smooth”:



Then, go under “Object Data Properties” → “Normals” and check “Auto Smooth” and change the angle to 40°:



Now, add a Bevel modifier, change the number of segments to 2, Angle to 40° and under “Shading” check “Harden Normals”:

